Personal copy · kamjunhao@gmail.com

```
import requests

base url = f"https://webexapis.com/v1/people"

headers = {
        "Accept": "application/json"
}

person email = 'customer@company.com'
url = f"{base url}?email={person email}"

response = requests.request("GET", url, headers=headers)

if response.status code == 200 :
        print(f"OK - Added {person email}")
else:
        print(f"ERROR CODE {response.status code}")
```

Refer to the exhibit. A collaboration engineer has developed a script to automate the gathering of information by providing the email address of the individual.

When the engineer tests the script, a 401 error is received. Which `command` fixes the script?

A Add ג€Authorizationג :€ג€Bearer <WEBEX_TOKEN>ג€ to the headers. ✓ CORRECT

B Add ג€Authenticationג :€ג€Basic <WEBEX_TOKEN>ג€ in the base_url after ג€https://ג€.

C Add ג€<Authorization>:<password>@ג€ in the base_url after ג€https://ג€.

D Add ג€Authenticationג :€ג€Bearer <WEBEX_TOKEN>ג€ to the headers.

SUGGESTED ANSWER

The suggested answer is A.

The correct command is to add 'Authorization': 'Bearer ' to the headers. This is because a 401 error indicates an unauthorized request, which suggests that the script is missing the necessary authorization credentials. Webex APIs typically require an OAuth token for authorization, and it should be included in the request headers with 'Bearer' followed by the

token. Therefore, adding this to the headers will properly authorize the request and fix the script.

QUESTION 100

A developer is attempting to retrieve all the messages from a Cisco Webex space. The server responds by sending the first 1,000 messages. The developer must make additional requests to retrieve the rest of the messages. Which API constraint is described in this scenario?

A payload limiting

B throttling

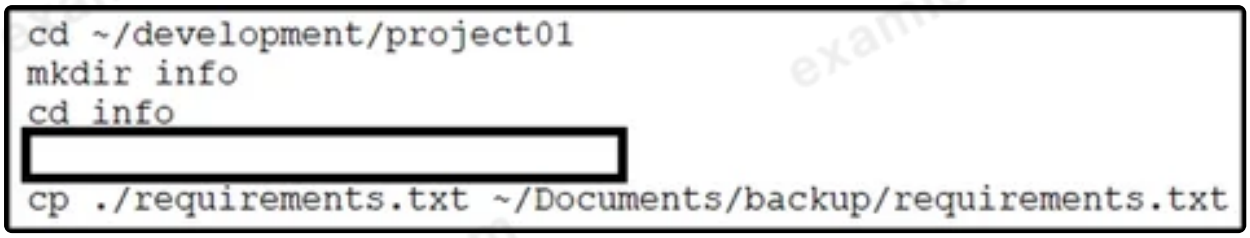
C pagination ✓ CORRECT

D rate limiting

SUGGESTED ANSWER

The suggested answer is C.

When a developer retrieves a large dataset from an API in chunks, rather than in a single request, it is referred to as pagination. This helps in efficiently fetching data without overloading servers or network resources. In this scenario, the server initially sends a subset of messages and the developer must request subsequent subsets, which is a classic example of pagination.



Refer to the exhibit. A developer just finished testing a Python script and wants to save the list of packages and versions installed on the current machine. The developer must make sure that it will not cause any issues if tested on another device due to different library versions. Which line of code needs to be placed on the snippet where the code is missing?

A pip freeze 2> requirements.txt

B pip freeze > requirements.txt ✓ CORRECT

C pip freeze => requirements.txt

D pip freeze | requirements.txt

SUGGESTED ANSWER

The suggested answer is B.

The correct command to save the list of installed Python packages and their versions to a file is 'pip freeze > requirements.txt'. The 'pip freeze' command outputs the installed packages, and the '>' operator redirects this output to the specified file, in this case, 'requirements.txt'.

Issued to kamjunhao@gmail.com · October 17, 2026

A developer pushes an application to production. The application receives a webhook over HTTPS without a secret. The webhook information contains credentials to service in cleartext. When the information is received, it is stored in the database with an SHA-256 hash. Credentials to the database are accessed at runtime through the use of a vault service. While troubleshooting, the developer sets the logging to **debug** to view the message from the webhook. What is the security issue in this scenario?

A Database credentials should be accessed by using environment variables defined at runtime.

B During the transport of webhook messages, the credentials could be unencrypted and leaked.

C During logging, debugging should be disabled for the webhook message. ✓ CORRECT

D Hashing the credentials in the database is not secure enough; the credentials should be encrypted.

SUGGESTED ANSWER

The suggested answer is C.

The main security issue here involves logging sensitive information. Although the transport security over HTTPS and the use of a hash for storing credentials are appropriate, enabling debug logging for the webhook message in a production environment introduces a significant vulnerability. This allows the credentials, which are received in plaintext, to be exposed through the debug logs. To mitigate this, debugging should be disabled, especially for sensitive information.

HTTP Request

```
curl http://hello-api.info -v
```

HTTP Response

```
*   Trying hello-app.info...
* TCP_NODELAY set
* Connected to hello-app.info (hello-app.info) port 80 (#0)
> GET / HTTP/1.1
> Host:hello-app.info
> User-Agent: curl/7.64.1
> Accept: */*
>

< HTTP/1.1 502 Bad Gateway
< Server: nginx/1.14.0 (Ubuntu)
< Date: Sat, 21 Nov 2020 11:09:54 GMT
< Content-Type: text/html
< Content-Length: 182
< Connection: keep-alive
```

Refer to the exhibit. A developer cannot reach the web application behind an NGINX load balancer. The developer sends a request to an application FQDN with cURL but gets an HTTP 502 response. Which action solves the problem?

A Fix errors in the server configuration, which is behind the load balancer. ✓ CORRECT

B Bring up the load balancer to the active state.

C Fix errors in the cURL request sent by the client.

D Change the default gateway on the load balancer to an active one.

SUGGESTED ANSWER

The suggested answer is A.

QUESTION 104

Issued to kamjunhao@gmail.com · October 17, 2026

```
import requests, sys
from config import user, password

url = "http://restconf.devnet.com:9443/api/running/interfaces"
headers = [redacted]
response = requests.put(url, auth=(user, password), headers=headers, verify=False)
print(response.text)
```

Refer to the exhibit. A network engineer wants use API update information about device interfaces. The network devices are deployed in a Cisco DevNet Sandbox and have running interfaces. Which value is suitable as the headers variable?

A {'Content-Type': 'application/yang.data+yaml', 'Accept': 'application/yang.data+yaml'}

B {'Content-Type': 'application/yang.data+json', 'Accept': 'application/yang.data+json'} ✓ CORRECT

C {'Content-Type': 'application/yang.data+utf8l', 'Accept': 'application/yang.data+utf8'}

D {'Content-Type': 'application/restconf.data+txt', 'Accept': 'application/restconf.data+txt'}

SUGGESTED ANSWER

The suggested answer is B.

The appropriate header for making API requests to RESTCONF interfaces is to specify the data format in which data is being sent and received. The value 'application/yang.data+json' indicates that the data will be in JSON format. RESTCONF manages data defined in YANG, and JSON (JavaScript Object Notation) is a commonly used format for representing structured data in web APIs. Therefore, the headers should be set to {'Content-Type': 'application/yang.data+json', 'Accept': 'application/yang.data+json'} to correctly specify that the data being sent and received is in JSON format.

Which action do webhooks enable an application to perform?

A Increase the request rate limit.

B Populate the application with restricted data.

C Receive real-time data. ✓ CORRECT

D Implement a push model.

SUGGESTED ANSWER

The suggested answer is C.

Webhooks enable an application to receive real-time data by allowing it to be notified immediately when certain events or activities occur. Instead of the application continuously checking (polling) for updates, webhooks provide a way for the application to receive updates automatically, thus ensuring that the data is received as soon as an event happens.

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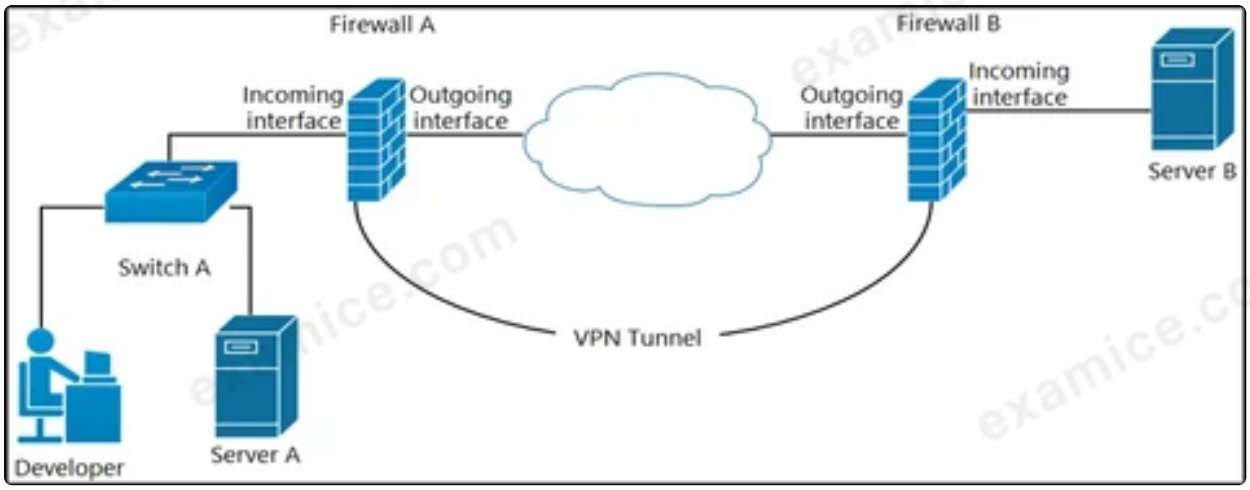
Issued to kamjunhao@gmail.com · October 17, 2026



Refer to the exhibit. A developer can access the TLS REST API on server A, but cannot access the API on server B. The developer can ping server B. When the developer performs a packet capture on the TLS REST API port on server B, the capture shows that the packet arrived and the server responded. What causes the issue?

A Port 80 is blocked on the outgoing interface of firewall B.

B Port 443 is blocked on the outgoing interface of firewall A. ✓ CORRECT

C Port 443 is blocked on the incoming interface of firewall B.

D Port 80 is blocked on the incoming interface of firewall A.

SUGGESTED ANSWER

The suggested answer is B.

The developer can ping server B, which means ICMP traffic is allowed through the network. The packet capture on server B shows that the packet arrives and the server responds, indicating that there is no issue with the incoming interface of firewall B. Therefore, the problem must be related to the outgoing traffic from server A through the VPN. Since TLS typically uses port 443 for secure connections, if port 443 is blocked on the outgoing interface of firewall A, the response packets would not be able to travel back through the VPN tunnel to the developer's machine, causing the issue.

DRAG DROP -

/api/access/global/rules API operations on global access rules. GET PUT PATCH POST DELETE

Implementation Notes

Add a new global access rule.

Parameters

Parameter	Required	Description	Type	Data Type
body	true	Definition of the global access rule to be added.	body	ExtendedACE

Response

Response Content Type application/json

Response Object ExtendedACE

ExtendedACE Model

Field	Value	Description	Constraints
sourceAddress*	object	source address in the ace	None
destinationAddress*	object	destination address in the ace	None
kind*	string	The kind of this resource object.	None
dstSecurity	object	destination security group in the ace	None
dstSecurity. Value	string		None
active*	boolean	shows if the rule is active or inactive, default is active	None
destinationService*	object	destination service in the ace	None
permit*	boolean	shows if the action is permit or deny	None

Refer to the exhibit. Drag and drop the code from the bottom onto the box where the code is missing to complete the function that adds a new global access rule that denies traffic using the Cisco ASA REST API. Not all options are used.

Select and Place:

```
import requests as req
def add rule(token, asa IP, dest v4Address):
    data = {
        "destinationAddress": {
            "kind": "IPv4Address",
            "value": dest v4Address
        },
        "destinationService": {
            "kind": "NetworkProtocol",
            "value": "ip"
        },
        "active": True,
        "sourceAddress": {
            "kind": "AnyIPAddress",
            "value": "any4"
        }
    }
    objectRequestUrl='https://'
    res = req.request(
                json=data,verify=False)
   print("Success.")
else:
   print("Failed")
```

```
if res.status_code == 201:
if res.status_code == 203:
+ asa_IP + "/api/access/global/rules"
'POST', objectRequestUrl,
"permit": False,
+ asa_IP = "/access/global/rules"
```

SUGGESTED ANSWER

```
import requests as req
def add rule(token, asa IP, dest v4Address):
    data = {
        "destinationAddress": {
            "kind": "IPv4Address",
            "value": dest v4Address
        },
        "destinationService": {
            "kind": "NetworkProtocol",
            "value": "ip"
        },
        "permit": False,
        "active": True,
        "sourceAddress": {
            "kind": "AnyIPAddress",
            "value": "any4"
        }
    }
    objectRequestUrl='https://' + asa_IP + "/api/access/global/rules"
    res = req.request( 'POST', objectRequestUrl,
                json=data,verify=False)
if res.status_code == 201:
   print("Success.")
else:
   print("Failed")
```

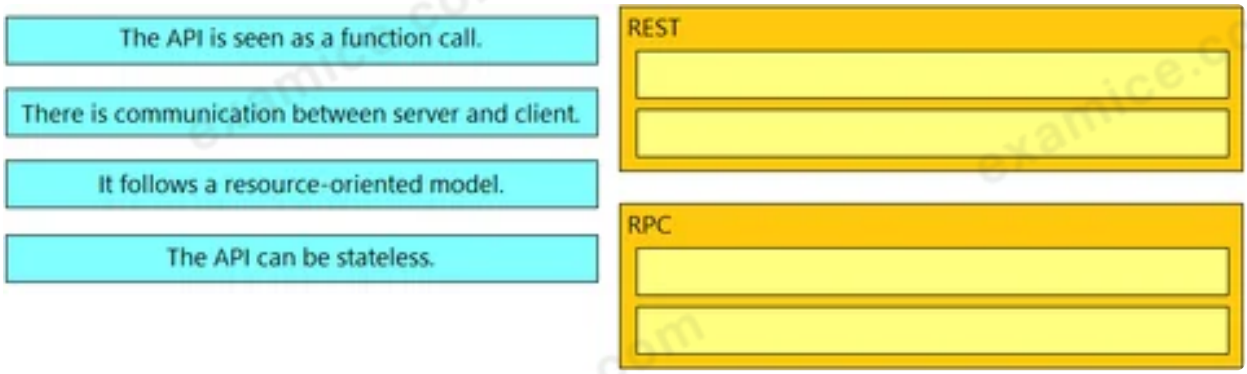
```
if res.status_code == 203:
```

```
+ asa_IP = "/access/global/rules"
```

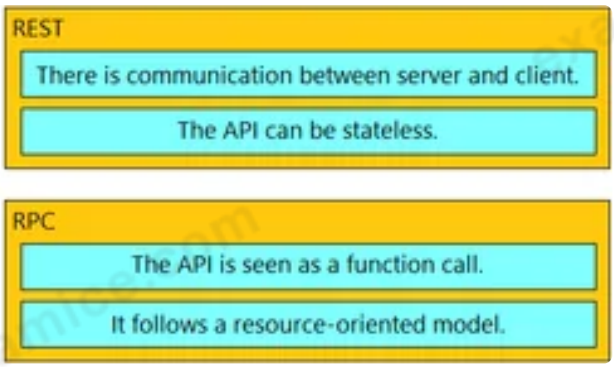
DRAG DROP -

Drag and drop the API characteristics from the left onto the categories on the right.

Select and Place:



SUGGESTED ANSWER



How are operations on REST APIs performed as compared to RPC APIs?

A In a REST API, operations are performed on an object (node) that is identified by a URL, but RPC APIs are operation-oriented. ✓ CORRECT

B In a REST API, operations are performed on an external resource that is defined on the API, but the RCP APIs are resource-oriented.

C In a REST API, operations are performed on a platform that is identified by the URL, but RPC APIs are platform-oriented.

D In a REST API, operations are performed on an internal resource that is defined on the API, but RPC APIs are resource-oriented.

SUGGESTED ANSWER

The suggested answer is A.

In a REST API, operations are performed on resources identified by URLs, utilizing HTTP methods such as GET, POST, PUT, and DELETE to manipulate these resources. This object-oriented approach contrasts with RPC APIs, which are operation-oriented, focusing on invoking specific functions or procedures rather than manipulating resources.

Issued to kamjunhao@gmail.com · October 17, 2026

Which HTTP error code series relates to redirection?

A 400

B 500

C 200

D 300 ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is D.

HTTP status codes are grouped into different classes based on the first digit of the code. The 3xx series of status codes indicates redirection. These codes inform the client that further action is needed to complete the request, such as moving to a different URL. Therefore, the 3xx series relates to redirection.

Issued to kamjunhao@gmail.com · October 17, 2026

Which platform has an API that can be used to obtain a list of vulnerable software on user devices?

A Cisco Umbrella

B Cisco Firepower

C Cisco Identity Services Engine

D Cisco Advanced Malware Protection ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is D.

Cisco Advanced Malware Protection (AMP) is designed to identify and provide information about vulnerable software present on user devices. AMP for Endpoints specifically includes capabilities to detect and list out vulnerable software, making it a suitable platform for obtaining such data via its API.

Issued to kamjunhao@gmail.com · October 17, 2026

Which two items are Cisco DevNet resources? (Choose two.)

A TAC Support

B Software Research

C API Documentation ✓ CORRECT

D Bitbucket

E Sandbox ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is C, E.

Cisco DevNet resources include API Documentation and Sandbox. API Documentation provides detailed information about the APIs that Cisco offers for developers, and Sandboxes are virtual environments where developers can experiment with Cisco technologies. These resources are designed to help developers create and test applications using Cisco products and technologies.

FILL BLANK -

Fill in the blanks to complete the Python script to update the Webex Teams membership of a room using the Python requests library. import requests url = `https://api.ciscospark.com/v1/memberships/Y2lzY29zcGFyazov379971079INISVAvOTJiM2RkOWEtN = `AbCdEf123456`payload = `{ \n \`isModerator\`: true\n}`headers = {'Authorization': 'Bearer' + my_token,'Content-Type': 'application/json'}response = requests. _____ (`PATCH`, url, headers= _____, data = _____) print(response.text.encode('utf8'))

SUGGESTED ANSWER

To update the Webex Teams membership of a room using the Python requests library, the correct approach is to use a PATCH request. In this case, the 'requests.request' method is a versatile way to achieve this as it allows specifying the HTTP method as a parameter. Hence, the correct way to fill in the blanks in the given Python script is as follows: response = requests.request('PATCH', url, headers= headers, data = payload) print(response.text.encode('utf8'))

Issued to kamjunhao@gmail.com · October 17, 2026

While developing a real-time VoIP application on a Cisco Webex platform, users report that their voice gets chopped or parts of the conversation drop out. Which network constraint is impacting the application?

A jitter ✓ CORRECT

B capacity

C delay

D latency

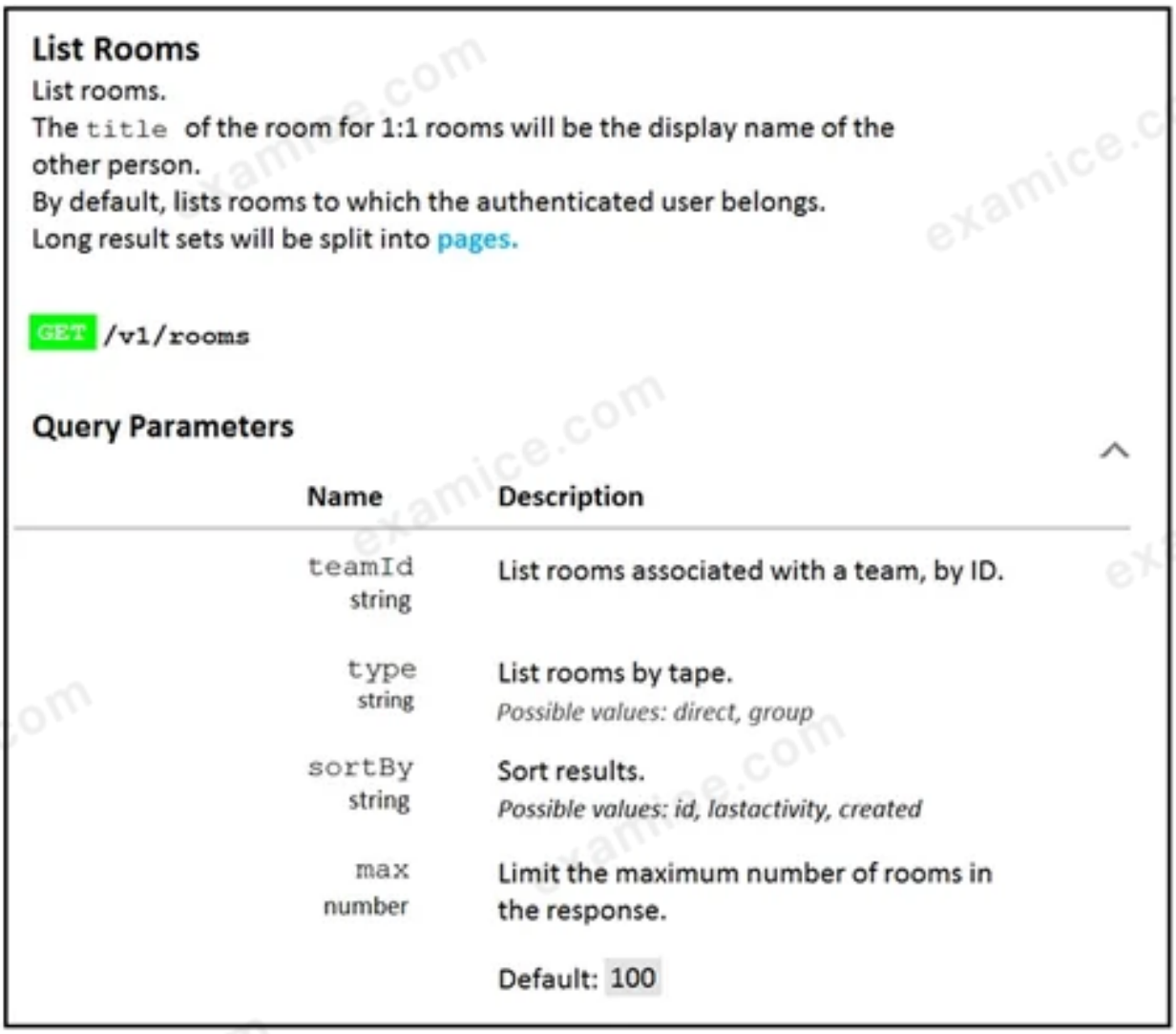
SUGGESTED ANSWER

The suggested answer is A.

The issue described, where the voice gets chopped or parts of the conversation drop out in a real-time VoIP application, is typically caused by jitter. Jitter refers to the variation in packet arrival times, and when it is high, it can cause significant problems for VoIP applications. High jitter can result in packets arriving out of sequence or late, which can cause the voice to sound choppy or parts of the conversation to be missing. This is because VoIP relies on a steady stream of data packets arriving in the correct order and on time to maintain clear audio quality. Therefore, jitter is the network constraint impacting the application.

FILL BLANK -

Refer to the exhibit.



List Rooms

List rooms.
The `title` of the room for 1:1 rooms will be the display name of the other person.
By default, lists rooms to which the authenticated user belongs.
Long result sets will be split into pages.

`GET` `/v1/rooms`

Query Parameters

Name	Description
`teamId` string	List rooms associated with a team, by ID.
`type` string	List rooms by tape. *Possible values: direct, group*
`sortBy` string	Sort results. *Possible values: id, lastactivity, created*
`max` number	Limit the maximum number of rooms in the response. Default: 100

Fill in the blank to complete the query parameter and value so that the result set is returned in reverse chronological order (most recent first) based on when messages were posted to those rooms.

HTTP GET /v1/rooms _____

SUGGESTED ANSWER

To return the result set in reverse chronological order (most recent first) based on when messages were posted to those rooms, you should use the 'sortBy' parameter with the value 'lastactivity'. Therefore, the complete query parameter and value should be: ?sortBy=lastactivity. So, the HTTP GET request will look like this: HTTP GET /v1/rooms?sortBy=lastactivity.

QUESTION 116

Issued to kamjunhao@gmail.com · October 17, 2026

A company has written a script that creates a log bundle from the Cisco DNA Center every day. The script runs without error and the log bundles are produced.

However, when the script is run during business hours, people report poor voice quality of phone calls. What explains this behavior?

A The script is written in a low-level programming language where there is no memory safety. This causes a buffer overflow and disruption on the network.

B The speed and duplex settings in Cisco DNA Center are set incorrectly, which causes the transfer to be too slow.

C The script is running in the Voice VLAN and causes delays and jitter in the subnet.

D Generating the logs causes the CPU on the network controller to spike, which causes delays in forwarding the voice IP packets. ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is D.

When the script runs and generates logs, it causes the CPU on the network controller to spike. This spike could lead to delays in processing tasks that are crucial for managing network resources and maintaining quality of service (QoS). Although the network controller is primarily involved with the control plane, CPU spikes can slow down the overall system's ability to manage network functions effectively, thereby causing poor voice quality as a result of delays in packet forwarding.

FILL BLANK -

Fill in the blanks to complete the statement.

Cisco DNA Center provides the capability to send an HTTP _____ request to the API endpoint https://DNA-C_API_ADDRESS/api/v1/ network-device/and receive a list of network _____ in _____ format.

SUGGESTED ANSWER

The correct answer is: 'GET', 'devices', 'JSON'. The HTTP method used to retrieve data from a server is typically 'GET'. The term 'devices' is correct as it matches the plural form used to indicate more than one network device. The format used is 'JSON', which stands for JavaScript Object Notation, commonly used for transmitting data in web applications. Therefore, the complete statement should be: Cisco DNA Center provides the capability to send an HTTP GET request to the API endpoint https://DNA-C_API_ADDRESS/api/v1/network-device/ and receive a list of network devices in JSON format.

FILL BLANK-

Fill in the blanks to complete the Python script to retrieve a list of network devices using the Cisco DNA Center API. my_token=

'eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.ey.JzdWIiOiI1ZDE0OWZkMjhlZTY2MmQ3NGM5YzE5ZTliL

import requestsurl = `https://myDNAserver/dna/intent/api/v1/network-device`payload = {}headers = {'x-auth-token': my_token}response = requests.request(_____ , url,headers = _____, data = _____) print(response.text.encode('utf8'))

SUGGESTED ANSWER

To retrieve a list of network devices using the Cisco DNA Center API, the appropriate method to use is 'GET'. The data for the GET request in this specific case can be an empty dictionary `{}`, and headers should include the authentication token. Therefore, the correct code to complete the script is: response = requests.request('GET', url, headers=headers, data=payload).

DRAG DROP -

Drag and drop the requests from the left into the order on the right to create and check the path trace between two devices using Cisco DNA center API.

Select and Place:

SUGGESTED ANSWER

DRAG DROP -

```
SDK Documentation:

Class: Devices

Device List: get_device_list()
Get device: get_device_by_id(id)
Delete device: delete_device_by_id(id)
Device status: inventoryStatusDetail
Device Parameters:
  id
  upTime
  type
```

Refer to the exhibit. A Python script must delete all Cisco Catalyst 9300 Series switches that have an uptime that is greater than 90 days. The script must also query for the status of all the other devices. Drag and drop the code from the bottom onto the box where the code is missing to complete the script. Not all options are used.

Select and Place:

Answer Area

```
from dnacentersdk import DNACenterAPI

device_type = "Cisco Catalyst 9300 Switch"
api_session = DNACenterAPI(
    base_url="https://sandboxdnac.cisco.com",
    username="user",password="password"
    )

devices = [                                                    ]

for device in devices:
    if int(device.upTime.split()[0]) > 90:
        if device.type == device type:
            output = [                                         ]
            print(output)

    else:
        selected device = [                                    ]
        output = [                                             ]
        print (output)
```

- `selected_device.response.inventoryStatusDetail(device)`
- `api_session.devices.delete_device_by_id(device.id)`
- `api_session.devices.get_device_by_id(device.id)`
- `selected_device.response.inventoryStatusDetail`
- `api_session.devices.get_device_list().response`

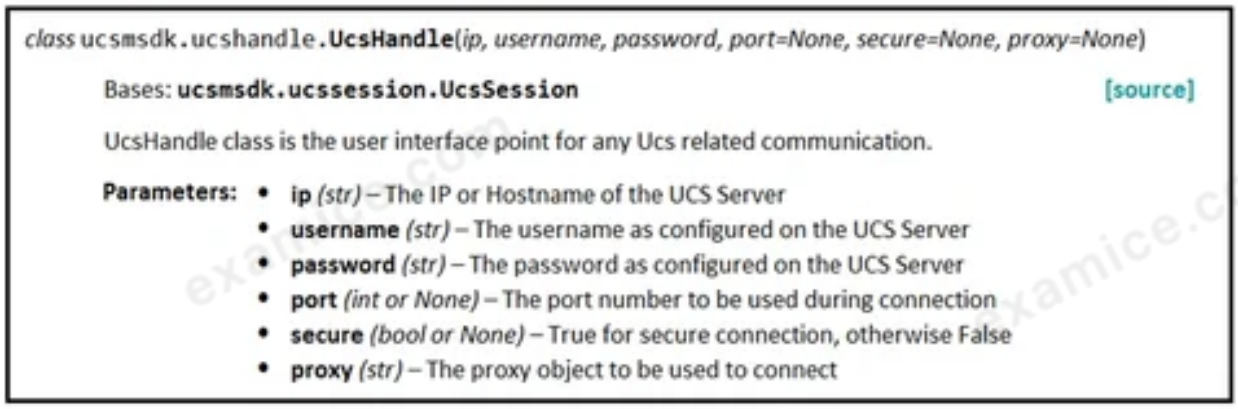
SUGGESTED ANSWER

Answer Area

```
from dnacentersdk import DNACenterAPI

device_type = "Cisco Catalyst 9300 Switch"
api_session = DNACenterAPI(
    base_url="https://sandboxdnac.cisco.com",
    username="user",password="password"
    )

devices = [api_session.devices.get_device_list().response]

for device in devices:
    if int(device.upTime.split()[0]) > 90:
        if device.type == device type:
            output = [api_session.devices.delete_device_by_id(device.id)]
            print(output)

    else:
        selected device = [api_session.devices.get_device_by_id(device.id)]
        output = [selected_device.response.inventoryStatusDetail]
        print (output)
```

```
selected_device.response.inventoryStatusDetail(device)
```

Refer to the exhibit.



class ucsmsdk.ucshandle.**UcsHandle**(*ip, username, password, port=None, secure=None, proxy=None*)

Bases: **ucsmsdk.ucssession.UcsSession** [source]

UcsHandle class is the user interface point for any Ucs related communication.

Parameters:
- **ip** *(str)* – The IP or Hostname of the UCS Server
- **username** *(str)* – The username as configured on the UCS Server
- **password** *(str)* – The password as configured on the UCS Server
- **port** *(int or None)* – The port number to be used during connection
- **secure** *(bool or None)* – True for secure connection, otherwise False
- **proxy** *(str)* – The proxy object to be used to connect

Given the API documentation for the UCS SDK Python class, UcsHandle, which code snippet creates a handle instance?

- A.

```
#!/usr/bin/env python3
from ucsmsdk.ucshandle import UcsHandle
handle = UcsHandle("10.1.2.254", "admin", "password",
   port="443", secure=True)
```

- B.

```
#!/usr/bin/env python3
from ucsmsdk.ucshandle import UcsHandle
handle = UcsHandle(ip="10.1.2.254", username="admin",
   password="password", port=443, secure=True)
```

- C.

```
#!/usr/bin/env python3
from ucsmsdk.ucshandle import UcsHandle
handle = UcsHandle("10.1.2.254", "admin", "password",
   port="443", secure=1)
```

- D.

```
#!/usr/bin/env python3
import ucsmsdk.ucshandle.UcsHandle
handle = UcsHandle("10.1.2.254", "admin", "password",
    port="443", secure=1)
```

SUGGESTED ANSWER

The given API documentation shows that the parameters for the `UcsHandle` class should be specified as follows: `ip`, `username`, `password`, `port`, `secure`, and `proxy`. The `port` parameter should be an integer, and the `secure` parameter should be a boolean (True/False). In the provided options, only option B correctly meets these criteria. It uses named parameters (which is acceptable in Python to improve readability), and the `port` parameter is an integer while the `secure` parameter is a boolean. Therefore, option B is correct.

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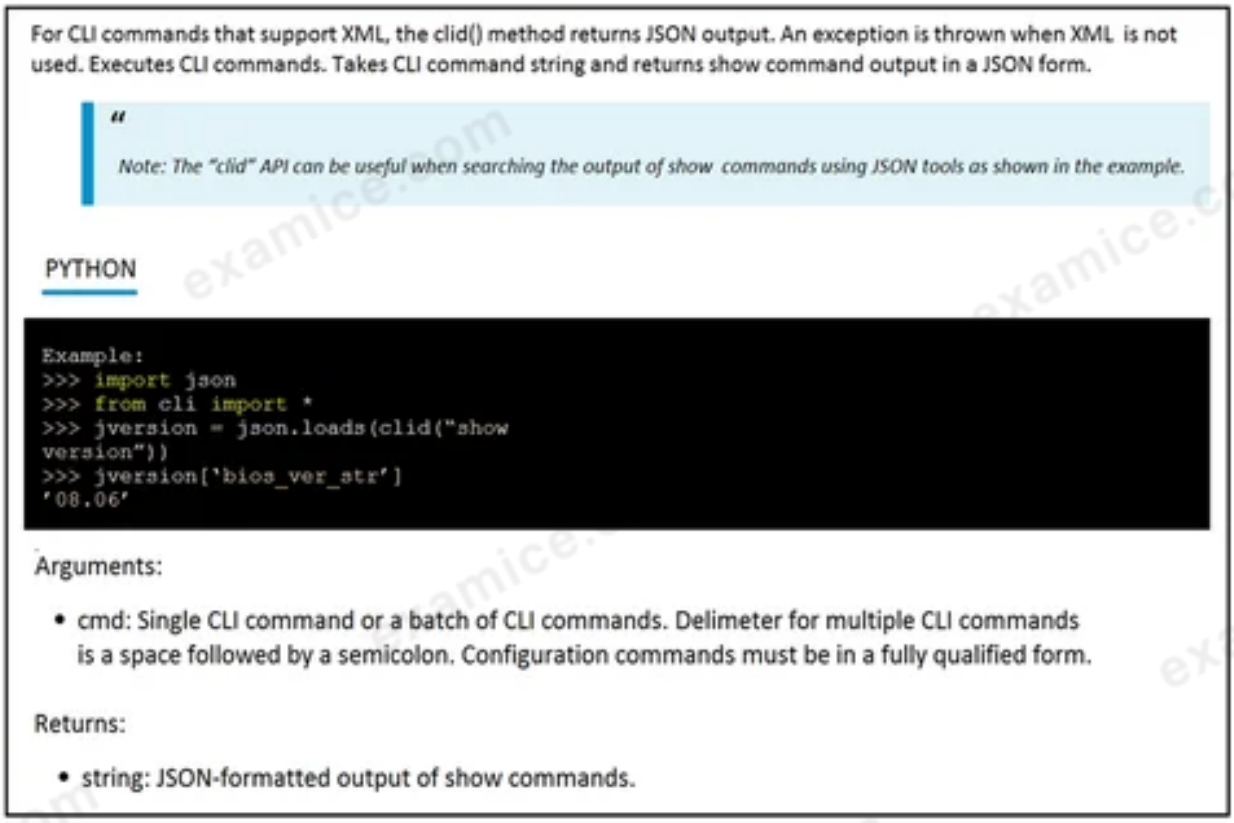
Refer to the exhibits.



For CLI commands that support XML, the clid() method returns JSON output. An exception is thrown when XML is not used. Executes CLI commands. Takes CLI command string and returns show command output in a JSON form.

> "
>
> *Note: The "clid" API can be useful when searching the output of show commands using JSON tools as shown in the example.*

PYTHON

```
Example:
>>> import json
>>> from cli import *
>>> jversion = json.loads(clid("show
version"))
>>> jversion['bios_ver_str']
'08.06'
```

Arguments:

- cmd: Single CLI command or a batch of CLI commands. Delimeter for multiple CLI commands is a space followed by a semicolon. Configuration commands must be in a fully qualified form.

Returns:

- string: JSON-formatted output of show commands.

```
>>> from cli import *
>>> import json

>>>
>>> cli('configure terminal ; interface loopback 5 ; no shut')
''
>>> intflist=json.loads(clid('show interface brief'))
>>> i=0
>>> while i < len (intflist['TABLE_interface']['ROW_interface']):
...     intf=intflist['TABLE_interface']['ROW_interface'][i]
...     i=i+1
...     if intf['state'] == 'up':
            print intf['interface']
```

The Python interpreter and the Cisco Python SDK are available by default in the Cisco NX-OS Software. The SDK documentation shows how the clid() API can be used when working with JSON and XML. What are two effects of running the script? (Choose two.)

A configure interface loopback 5 ✓ CORRECT

B show details for the TABLE interface

C issue shutdown on interface loopback 5

D show only the interfaces in the up status ✓ CORRECT

E show only the interfaces in admin shut status

SUGGESTED ANSWER

The suggested answer is A, D.

The provided script first issues a CLI command to configure an interface, specifically 'interface loopback 5', and then brings it up by using 'no shut'. This configuration command is represented in the statement 'cli('configure terminal ; interface loopback 5 ; no shut')'. Therefore, one effect of the script is to configure 'interface loopback 5'. Next, the script captures the output of 'show interface brief' and loads the components into a JSON object. It then iterates through this object to find interfaces that are in the 'up' state and prints these interfaces. Thus, another effect of the script is showing only the interfaces that are in the 'up' status. Therefore, the correct answers are configuring 'interface loopback 5' and showing only the interfaces in the 'up' status.

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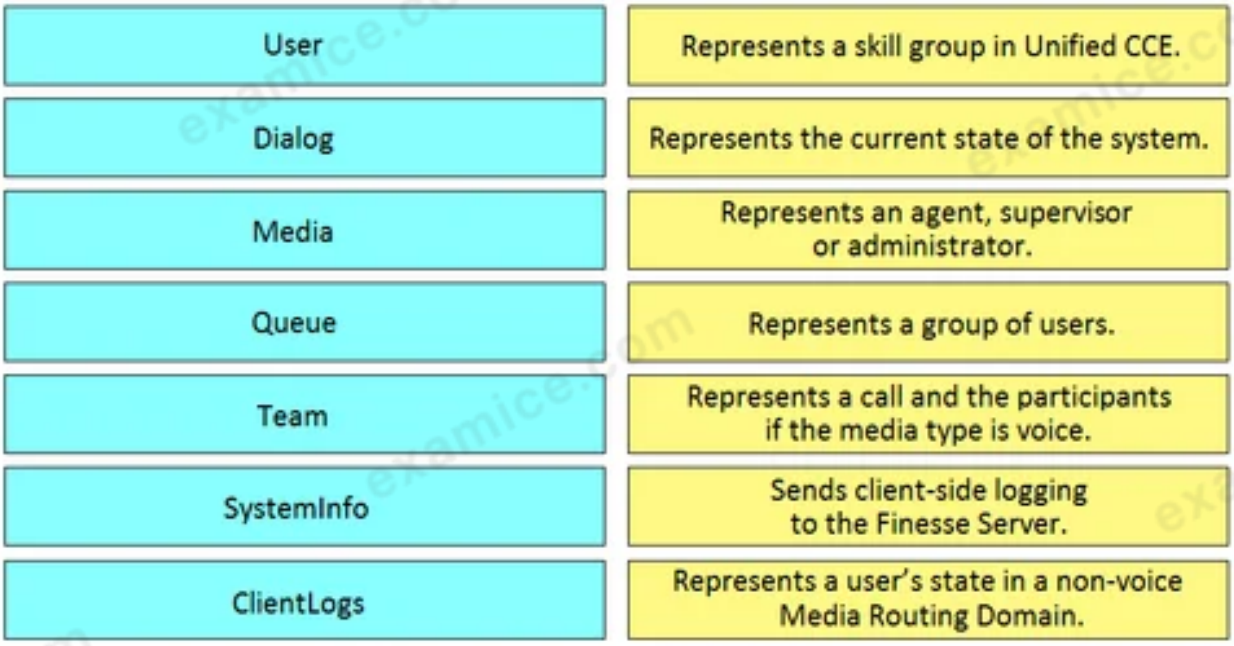
DRAG DROP -

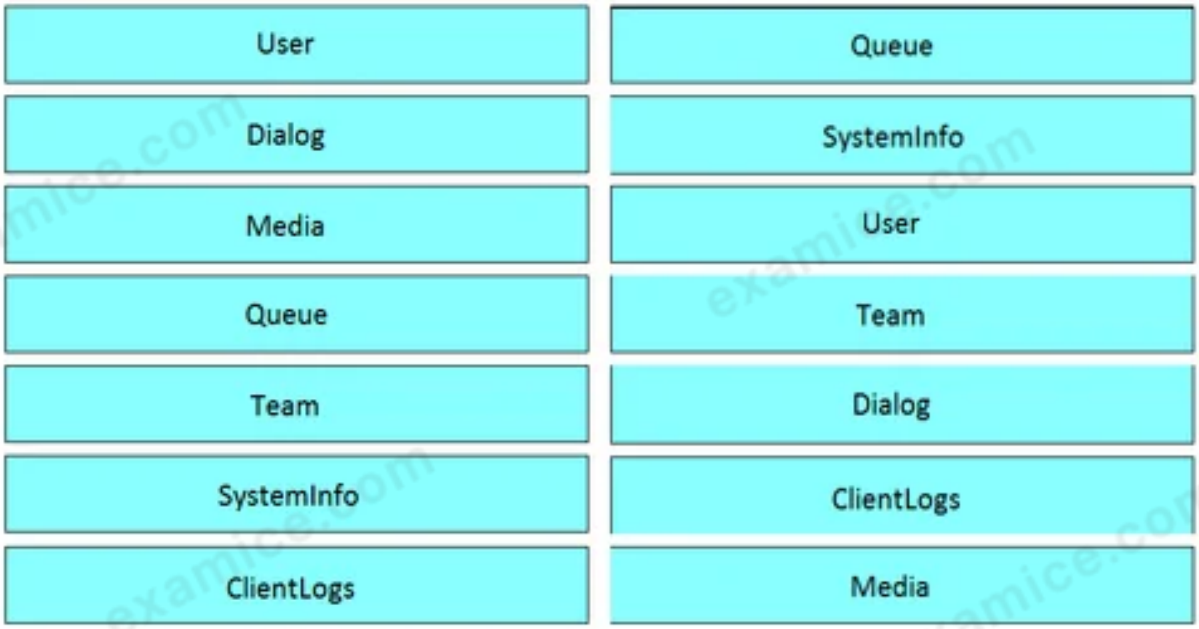
Drag and drop the element of the Finesse API on the left onto its functionality on the right.

Select and Place:



SUGGESTED ANSWER

Reference:
https://developer.cisco.com/docs/finesse/#!cisco-finesse-desktop-apis



QUESTION 124

Issued to kamjunhao@gmail.com · October 17, 2026

What are two key capabilities of Cisco Finesse? (Choose two.)

A Agents access Finesse from a browser without needing to install or configure anything on the client machine. ✓ CORRECT

B An OpenDNS utility is preconfigured and ready to use on Finesse.

C Gadget containers provide a seamless experience in a single user interface. ✓ CORRECT

D Finesse automatically collects telemetry data.

E Finesse includes an RPC API that enables the development of custom gadgets.

SUGGESTED ANSWER

The suggested answer is A, C.

Agents can access Cisco Finesse from a browser without needing to install or configure anything on the client machine, making it easy to use and deploy. Additionally, gadget containers in Cisco Finesse provide a seamless experience in a single user interface, allowing for the integration of various small web-based applications, known as gadgets, into the Finesse UI.

Which two statements are

true about Cisco UCS Manager, Cisco UCS Director, or Cisco Intersight APIs? (Choose two.)

A UCS Manager uses JSON to encode API interactions and utilizes Base64-encoded credentials in the HTTP header for authentication.

B UCS Director API interactions can be XML- or JSON-encoded and require an API key in the HTTP header for authentication. ✓ CORRECT

C Cisco Intersight uses XML to encode API interactions and requires an API key pair for authentication.

D UCS Manager API interactions are XML-encoded and require a cookie in the method for authentication. ✓ CORRECT

E Cisco Intersight API interactions can be encoded in XML or JSON and require an API key in the HTTP header for authentication.

SUGGESTED ANSWER

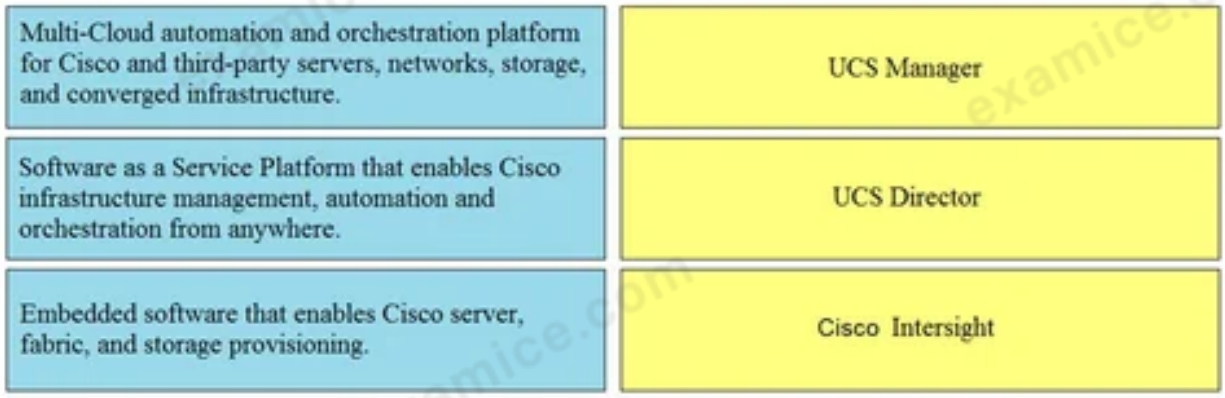
The suggested answer is B, D.

UCS Director API interactions can be encoded in XML or JSON and typically require an API key in the HTTP header for authentication. UCS Manager, however, uses XML to encode its API interactions and requires a cookie for authentication. These statements accurately describe how these systems manage API encoding and authentication mechanisms.

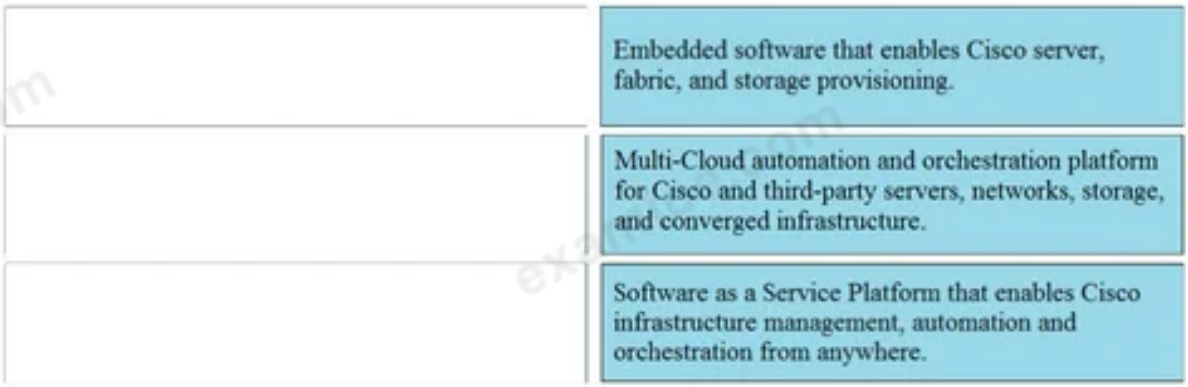
DRAG DROP -

Drag and drop the capability on the left onto the Cisco compute management platform that supports the capability on the right.

Select and Place:



SUGGESTED ANSWER



Issued to kamjunhao@gmail.com · October 17, 2026

Which model-driven programmability protocol does Cisco IOS XE Software support?

A gNMI ✓ CORRECT

B SOAP

C SSH

D CORBA

SUGGESTED ANSWER

The suggested answer is A.

Cisco IOS XE Software supports the gNMI (gRPC Network Management Interface) protocol for model-driven programmability. The gNMI protocol allows for robust and flexible network management and configuration by utilizing data models for streamlined operations.

Issued to kamjunhao@gmail.com · October 17, 2026

Which operation is performed with YANG model-driven programmability in NX-OS?

A configure a device with native and OpenConfig-based models ✓ CORRECT

B bootstrap a device that has a factory-default configuration

C send CLI commands to a device and retrieve output in JSON format

D run Linux commands natively on the device

SUGGESTED ANSWER

The suggested answer is A.

YANG model-driven programmability in NX-OS allows the configuration of a device using both native and OpenConfig-based models. YANG is a data modeling language used to model configuration and state data manipulated by NETCONF. It provides a standardized way to represent configuration and operational data, and OpenConfig is a notable industry initiative for defining vendor-neutral YANG models. Therefore, the correct operation performed with YANG model-driven programmability in NX-OS is configuring a device with native and OpenConfig-based models.

Issued to kamjunhao@gmail.com · October 17, 2026

What are two benefits of model-driven programmability? (Choose two.)

A easier to design, deploy, and manage APIs

B single choice of transport, protocol, and encoding

C models decoupled from transport, protocol, and encoding ✓ CORRECT

D model-based, structured, and human friendly ✓ CORRECT

E model-driven APIs for abstraction and simplification

SUGGESTED ANSWER

The suggested answer is C, D.

Model-driven programmability provides significant benefits, including being model-based, structured, and human-friendly. This makes it easier for developers to understand and work with the models, improving efficiency and reducing errors. Additionally, it decouples models from transport, protocol, and encoding, allowing for greater flexibility and adaptability in various environments. These benefits are essential for maintaining a robust, scalable, and efficient programmable infrastructure.

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What is a benefit of model-driven programmability?

A. It provides a programmatic and standards-based way of writing configurations to any network device. ✓ CORRECT

B. By modeling data, it ensures that vast quantities of data are usable for monitoring.

C. It simplifies management and operations by decoupling the networking hardware from its control mechanism.

D. It addresses the static architecture of traditional networks being decentralized and complex.

SUGGESTED ANSWER

The suggested answer is A.

A benefit of model-driven programmability is that it provides a programmatic and standards-based way of writing configurations to any network device. This method allows configurations to be described in a standardized language, ensuring compatibility and easing the configuration process across different types of network devices. This approach eliminates the need for device-specific commands or scripts, simplifying the management and enhancing interoperability.

Issued to kamjunhao@gmail.com · October 17, 2026

Which type of HTTP method is used by the Meraki and Webex Teams APIs to send webhook notifications?

A HTTP POST ✓ CORRECT

B HTTP GET

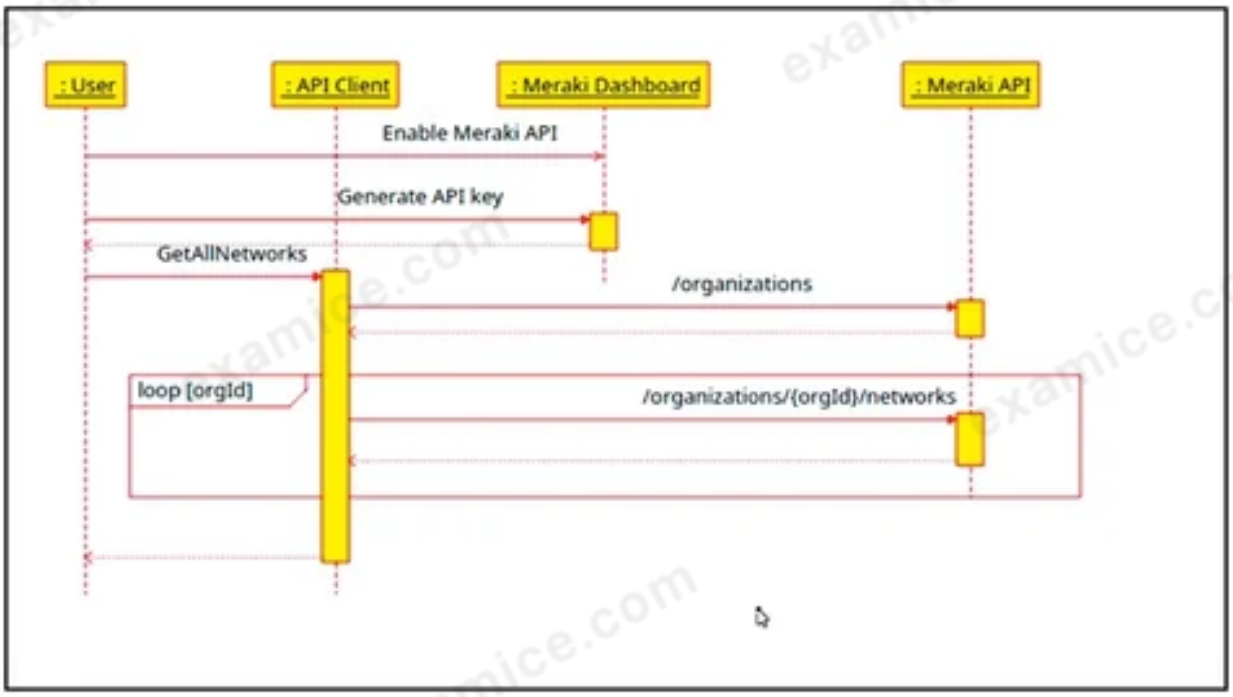
C HTTP HEAD

D HTTP PUT

SUGGESTED ANSWER

The suggested answer is A.

The HTTP POST method is commonly used for webhook notifications as it allows the server to send a message body to the specified URL, notifying the app of an event or activity. This is the standard method for sending data in webhook implementations by APIs, including Meraki and Webex Teams.



Refer to the exhibit. The Cisco Meraki API manages two organizations. In each organization, two independent networks are defined. How many calls to the / organizations/{orgId}/networks endpoint will be run parallel by the GetAllNetworks function of the API client?

A 0

B 1 ✓ CORRECT

C 2

D 4

SUGGESTED ANSWER

The suggested answer is B.

The number of calls to the /organizations/{orgId}/networks endpoint that will be run in parallel by the GetAllNetworks function of the API client is 1. This is because the synchronous operations are organized in a loop, iterating over each organization (orgId) sequentially. Given that the solid arrowheads and arrows with dotted lines signify synchronous calls and responses, it confirms that only one call is processed at any given moment. Therefore, only one call to the endpoint will run in parallel by the API client.

Refer to the exhibit.

```
$ pyang -f tree Cisco-IOS-XE-native.yang
module: Cisco-IOS-XE-native
  +--rw native
     +--rw logging
        +--rw monitor!
           +--rw severity?  logging-level-type

$ curl --insecure -v -X PATCH \
     "https://ios-xe-mgmt-latest.cisco.com:9443/restconf/data/Cisco-IOS-XE-native:native" \
     -H 'Authorization: Basic ZGV2ZWKvCGVyOkMXc2NVMTIzNOU=' -H 'Accept: application/yang-data+json' \
     -H 'Content-Type: application/yang-data+json' \
     -d '{ "Cisco-IOS-XE-native:native" : { "logging" : { "monitor" : { "severity" : "alerts" }} } }'
> ...
< ...
< HTTP/1.1 204 No Content
```

The definition of the YANG module and a RESTCONF query sent to a Cisco IOS XE device is shown. Why does the RESTCONF interface respond with a **204** status code?

A The query retrieved the logging severity level.

B The query updated the logging severity level or created it if not present on the device. ✓ CORRECT

C The query failed at retrieving the logging severity level.

D The query failed to set the logging severity level due to an invalid path.

SUGGESTED ANSWER

The suggested answer is B.

A 204 status code indicates that the server successfully processed the request and is not returning any content. In the context of the RESTCONF query shown, the use of the PATCH method signifies a request to update a resource or create it if it does not already exist. Given that the logging severity level is being specified in the request, the server would return a 204 status code to indicate that this update operation was completed successfully.

Issued to kamjunhao@gmail.com · October 17, 2026

What is a requirement when authenticating a RESTCONF API session with a router that runs Cisco IOS XE Software?

A No authentication is required.

B OAuth must be used.

C A token must be retrieved and the associated string must be embedded in the X-Auth-Token header.

D Basic authentication must be used. ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is D.

When authenticating a RESTCONF API session with a router that runs Cisco IOS XE Software, basic authentication must be used. This involves using typical HTTP authentication models where usernames and passwords are Base64 encoded and transmitted from the client to the server via an Authentication header.

Issued to kamjunhao@gmail.com · October 17, 2026

Which API **must** an engineer use to change a netmask on a Cisco IOS XE device?

A Meraki

B SD-WAN

C RESTCONF/YANG ✓ CORRECT

D DNAC

SUGGESTED ANSWER

The suggested answer is C.

To change a netmask on a Cisco IOS XE device, an engineer should use RESTCONF/YANG. RESTCONF is a protocol used for accessing data defined in YANG using REST-like APIs. This combination allows for network management tasks including configuration changes such as updating a netmask.

Issued to kamjunhao@gmail.com · October 17, 2026

A developer creates a script that configured multiple Cisco IOS XE devices in a corporate infrastructure. The internal test environment is unavailable, and no maintenance window is available to test on a low-priority production environment. Which resource is used to test the code before it is applied to the production environment?

A Code Exchange

B Cisco Support

C Cisco DevNet Learning Labs

D Cisco DevNet Sandbox ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is D.

When physical access to a test environment is unavailable and testing on a low-priority production system is not an option, a virtualized test environment can be used. Cisco DevNet Sandbox provides a fully virtualized environment that allows developers to test their scripts, configurations, and applications on Cisco platforms without the risk of impacting production systems. It is designed specifically for such scenarios and offers a range of pre-built environments that can be accessed remotely.

Which tool is used to block all traffic to the domain by using a single API call?

A Cisco Firepower

B Cisco AMP

C Cisco Umbrella ✓ CORRECT

D Cisco ISE

SUGGESTED ANSWER

The suggested answer is C.

Cisco Umbrella is the appropriate tool for blocking all traffic to a domain using a single API call. It leverages DNS-based security to prevent client connections by blocking the resolution of DNS for the specified domain. This effectively stops the client from connecting to and potentially downloading malicious content from the domain.

examice.com

Issued to kamjunhao@gmail.com · October 17, 2026

Why is refactoring done during test-driven development?

A to enable larger objects and longer methods

B to improve maintainability and readability ✓ CORRECT

C to ensure that the previous uses of the objects are explained

D to ensure the duplication of essential code

SUGGESTED ANSWER

The suggested answer is B.

Refactoring in test-driven development (TDD) is primarily done to improve the maintainability and readability of code. During TDD, developers write tests before writing the actual code. Once the code is written and passes the tests, refactoring is carried out to clean up the code, making it more efficient, understandable, and easier to maintain in the future. This practice ensures that the code base remains in good shape while allowing new features to be added easily.

```
from requests import request
from requests.auth import HTTPBasicAuth
from sys import argv
url = "https://"+ argv[1] +":443/restconf/data/native/ip/route"
auth = HTTPBasicAuth('developer', 'Cisco12345')
payload = {
    "Cisco-IOS-XE-native:route": {
        "ip-route-interface-forwarding-list": [
            {
              "prefix": argv[2],
              "mask": argv[3],
              "fwd-list": [
                  {
                      "fwd": "GigabitEthernet2",
                      "interface-next-hop": [
                          {
                              "ip-address": argv[4]
                          }
                      ]
                  }
              ]
          }
      ]
    }
  }
headers = {
    'Content-Type': 'application/yang-data+json',
    'Accept': 'application/yang-data+json',
}
response = request("PATCH", url, auth=auth, headers=headers, json = payload,
verify=False)
```

Refer to the exhibit. A network engineer must change the configuration on devices daily. The `Python` script must collect CLI arguments from the user, organize data according to the `Cisco-IOS-XE-native` YANG model, and utilize Cisco REST APIs to make the configuration. Which workflow is being automated by the

A updating an existing route on the device

B adding a new route to a device ✓ CORRECT

C including a route on the device and overwriting the current routes

D deleting the route that is passed through the command-line variables

SUGGESTED ANSWER

The suggested answer is B.

The Python script is designed to collect command-line arguments from the user, organize data according to the specified YANG model, and utilize Cisco REST APIs to make configurations on a device. The use of the PATCH method in this context indicates that the script is intended to add a new route to the device. The PATCH method allows for modifications to specific parts of a resource, and if the resource does not exist, it can create it. Thus, it aligns with the workflow of adding a new route rather than updating or deleting an existing one.

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QUESTION 140

Issued to kamjunhao@gmail.com · October 17, 2026

Which platform is used to programmatically create a space and invite users for collaboration?

A Cisco Intersight

B Cisco Finesse

C Cisco Webex ✓ CORRECT

D Cisco UCM

SUGGESTED ANSWER

The suggested answer is C.

Cisco Webex is the platform used to programmatically create a space and invite users for collaboration. It provides APIs and tools that allow developers to create spaces, invite participants, and facilitate online meetings and teamwork. Cisco Intersight, Cisco Finesse, and Cisco UCM do not have features primarily designed for creating collaboration spaces and managing user invitations in the same way.

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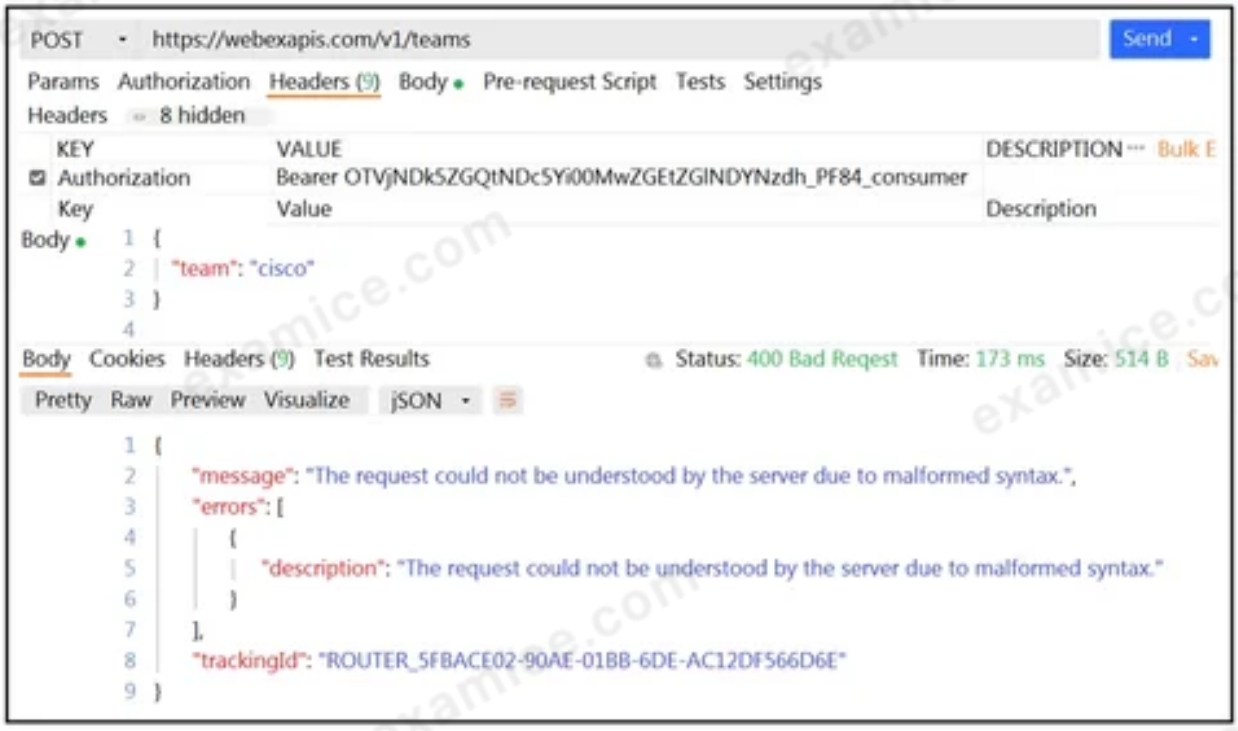


Refer to the exhibit. A developer wants to create a room named cisco through Webex API. The developer prepares the request by putting an API token inside the authorization header; however, the given HTTP call returns error code 400. How must the call parameters be changed to ensure a valid response?

A Replace the team body parameter with the name body parameter. ✓ CORRECT

B Replace cisco with a valid name for a team.

C Change the token in the header with the authorization token.

D Change the HTTP method of the call to PUT.

SUGGESTED ANSWER

The suggested answer is A.

The error code 400 indicates a syntax error in the request parameters. In the Webex API, when creating a room, the correct parameter to specify the room's name is 'title', not 'team'. Therefore, replacing the 'team' body parameter with 'title' will correct the syntax of the request and ensure a valid response.

QUESTION 142

Issued to kamjunhao@gmail.com · October 17, 2026

```
$ sh script.sh
Traceback (most recent call last):
  File "script.py", line 2, in <module>
    api = WebexTeamsAPI()
  File "/opt/.venv/pyvenv35/lib/python3.5/site-packages/webexteamssdk/api/__init__.py", line 112,
in __init__
    "You must provide a Webex Teams access token to interact with "
webexteamssdk.exceptions.AccessTokenError: You must provide a Webex Teams access token to interact
with the Webex Teams APIs, either via a WEBEX_TEAMS_ACCESS_TOKEN environment variable or via the
access_token argument.
$
$
$ cat script.sh
#!/bin/bash

python script.py
$
```

Refer to the exhibit. Which line **must** be included in the Bash script for the Python code to succeed?

A define WEBEX_TEAMS_ACCESS_TOKEN=your_access_token_here

B export WEBEX_TEAMS_ACCESS_TOKEN=your_access_token_here ✓ CORRECT

C set WEBEX_TEAMS_ACCESS_TOKEN=your_access_token_here

D WEBEX_TEAMS_ACCESS_TOKEN=your_access_token_here

SUGGESTED ANSWER

The suggested answer is B.

To run a Python script that requires a Webex Teams access token, you must set the environment variable correctly in the shell script. The appropriate way to set an environment variable in a Bash script in a Unix-like system is using the export command. The line 'export WEBEX_TEAMS_ACCESS_TOKEN=your_access_token_here' will set the environment variable so that it is available to the Python script when it runs.

Issued to kamjunhao@gmail.com · October 17, 2026

What is a capability of the NETCONF protocol?

A allows Python scripts to be executed externally from a network device

B uses JSON to encode data in its content layer

C allows XML messages to be exchanged via the SSH protocol ✓ CORRECT

D exposes a web-based interface

SUGGESTED ANSWER

The suggested answer is C.

NETCONF (Network Configuration Protocol) allows XML messages to be exchanged via the SSH protocol. This is a key capability of NETCONF, which enables secure communication and configuration between network devices using a standardized protocol.

```
<rpc message-id="101" xmlns="urn:ietf:params:xml:ns:Netconf:base:1.0">
  <get-conflg>
    <source>
      <running/>
    </source>
    <filter>
      <interfaces xmlns=" urn:ietf:params:xml:ns:yang:ietf-interfaces">
        <interface>
          <name/>
        </interface>
      </interfaces>
    </filter>
  </get-config>
</rpc>
```

Refer to the exhibit. A network engineer uses model-driven programmability to monitor and perform changes on the network. The network engineer decides to use the NETCONF RPC message to complete one of their tasks. What is accomplished by sending the RPC message?

A The running-config of the device is returned.

B The name of each interface is reset to a default name.

C All the YANG capabilities supported by the device are returned.

D A list of interface names is returned. ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is D.

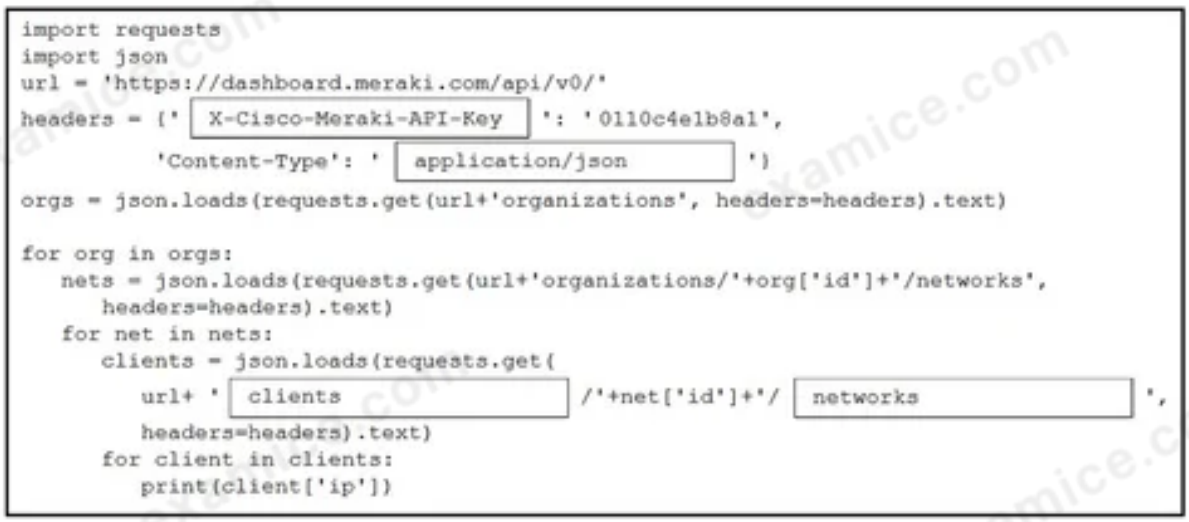
The NETCONF RPC message shown includes a filter for interfaces and specifies that the running configuration is the source. This means that the request is asking for specific information within the running configuration, filtered by interfaces. The filter further specifies that it is interested in the interface names. Therefore, the RPC message will return a list of interface names.

DRAG DROP -

Drag and drop the code from the bottom onto the box where the code is missing in the Python script to complete a function to return all the clients on a network by using the Cisco Meraki Dashboard API. Not all options are used.

Select and Place:

SUGGESTED ANSWER



client

application/xml

Authorization: Bearer

```
import requests
from requests.auth import HTTPBasicAuth
from dnac_config import DNAC, DNAC_PORT, DNAC_USER, DNAC_PASSWORD

def get_auth_token():
    url = 'https://sandboxdnac.cisco.com/dna/system/api/v1/auth/token'
    # API Endpoint: requires POST method
    [                                                    ]

def get_device_list():
    """
    Building out function to retrieve list of devices. Using requests.get to make a call
    to the network de-vice Endpoint
    """
    token = get_auth_token() # Get Token
    url = 'https://sandboxdnac.cisco.com/api/v1/network-device'
    hdr = {'x-auth-token': token, 'content-type' : 'application/json'}
    resp = requests.get(url, headers=hdr)
    device_list = resp.json()
    print_device_list(device_list)

if __name__ == "__main__":
    get_device_list()
```

Refer to the exhibit. A developer creates a script to obtain a list of devices by using the Cisco DNA Center API. The remote server authorizes the request only if an authentication token is supplied in the headers. A function named get_auth_token() must retrieve a valid token by using HTTP Basic Authentication. Which code must be added to complete the get_auth_token() function?

A resp = requests.post(url, auth=HTTPBasicAuth(DNAC_USER, DNAC_PASSWORD)) token = resp.json()['Token'] return token ✓ CORRECT

B resp = requests.post(url, auth=(DNAC_USER, DNAC_PASSWORD)) token = resp.json () ['Token'] return token

C resp = http.post(url, auth=HTTPBasicAuth(DNAC_USER, DNAC_PASSWORD)) token = resp.json()['Token'] return token

D resp = http.post(url, auth=(DNAC_USER, DNAC_PASSWORD)) token = resp.json() ['Token'] return token

SUGGESTED ANSWER

The suggested answer is A.

> The correct code that must be added to complete the get_auth_token() function involves making a POST request using the requests module, and using the HTTPBasicAuth method for basic authentication. This ensures that the username and password are properly encoded and sent in the request header. Furthermore, the response is parsed as JSON to extract the token. Therefore, the correct code snippet is: resp = requests.post(url, auth=HTTPBasicAuth(DNAC_USER, DNAC_PASSWORD)) token = resp.json()['Token'] return token.

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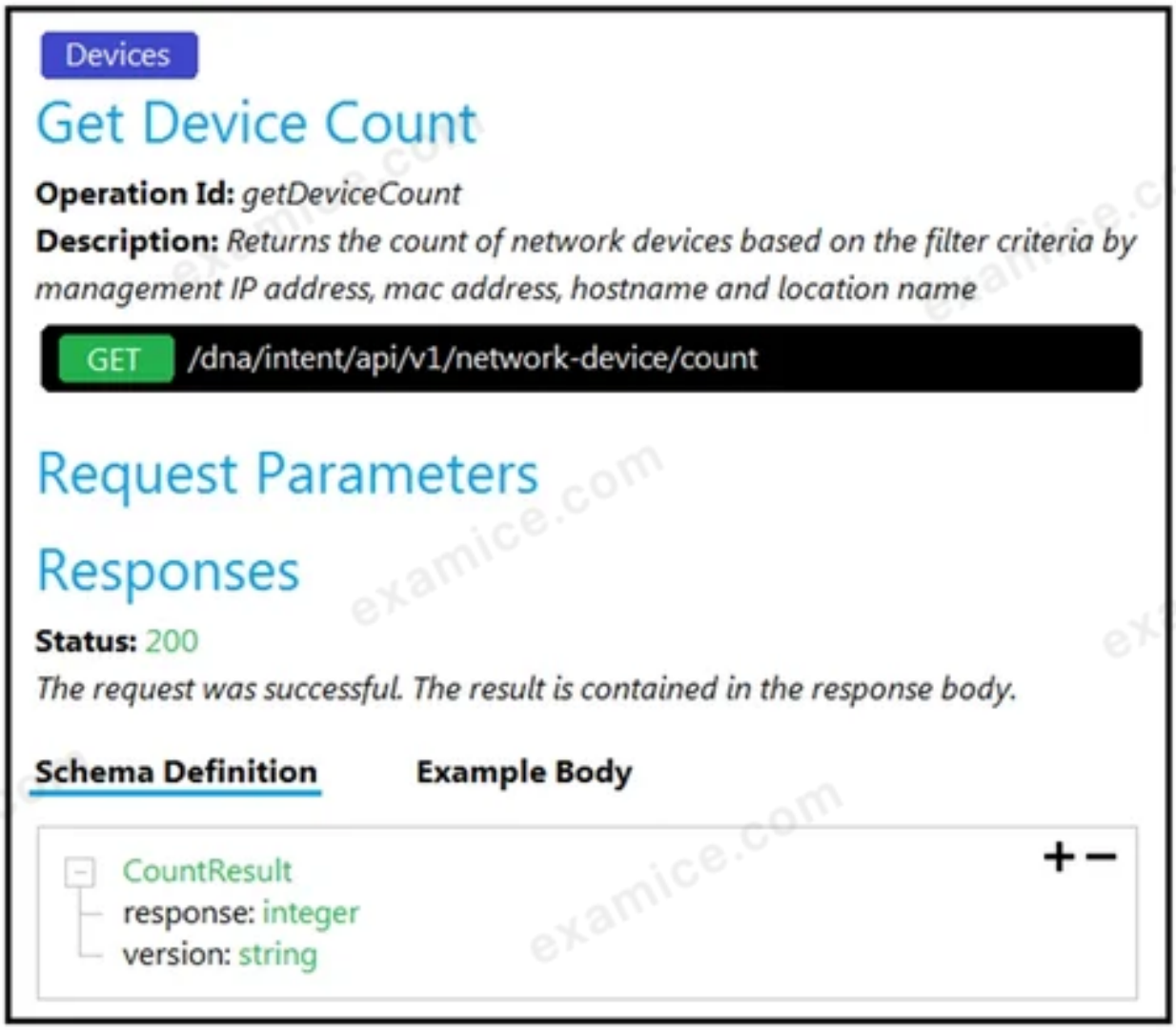
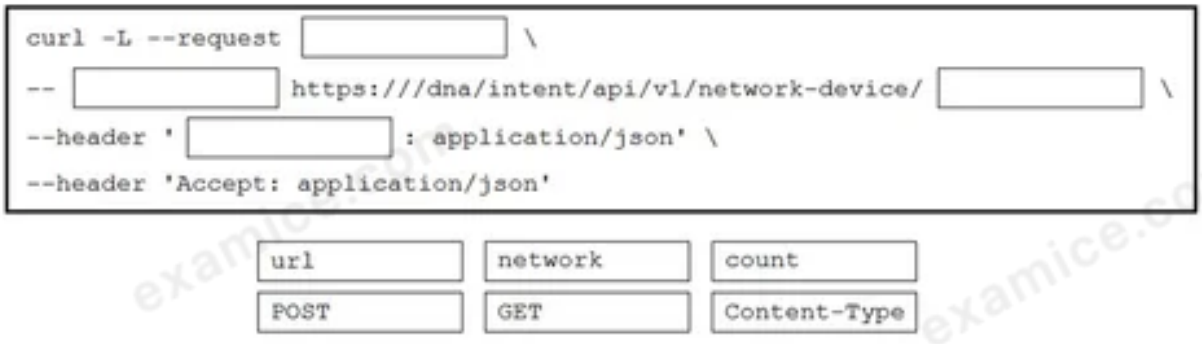
DRAG DROP -

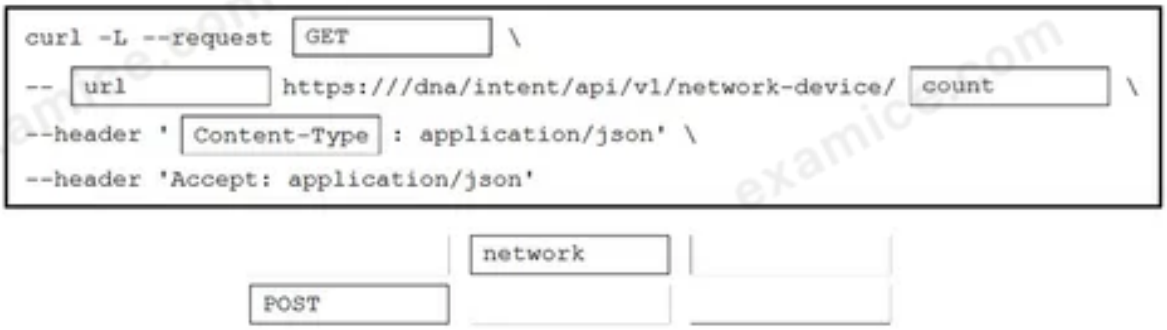


Refer to the exhibit. Drag and drop the code snippets from the bottom onto the blanks in the code to construct a request to find the Cisco DNA Center switch count. Not all options are used.

Select and Place:



SUGGESTED ANSWER



network

POST

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```
import requests, sys
from config import user, password

url = "http://restconf.devnet.com:9443/api/running/interfaces"
headers = 
response = requests.put(url, auth=(user, password), headers=headers,
verify=False)
print(response.text)
```

Refer to the exhibit. A network engineer wants to use an API to update information about device interfaces. The network devices are deployed in a Cisco DevNet Sandbox and have running interfaces. Which line of code needs to be placed in the snippet where the code is missing?

A {'Content-Type': 'application/yang.data+xml', 'Accept': 'application/yang.data+xml'} ✓ CORRECT

B {'Content-Type': 'application/yang.data+yaml', 'Accept': 'application/yang.data+yaml'}

C {'Content-Type': 'application/restconf.data+json', 'Accept': 'application/restconf.data+json'}

D {'Content-Type': 'application/yang.data+utf8', 'Accept': 'application/restconf.data+utf8'}

SUGGESTED ANSWER

The suggested answer is A.

In the context of RESTCONF, the correct MIME type for data transmitted using XML format is 'application/yang.data+xml'. This MIME type indicates that the data follows the YANG data model and is in XML format. The line of code that would specify these headers correctly is {'Content-Type': 'application/yang.data+xml', 'Accept': 'application/yang.data+xml'}.

```
#/bin/bash

curl https://ciscodevnet.com/devnet-0.13.tar.gz -o /tmp/devnet.tar.gz
tar -zxf /tmp/devnet.tar.gz
mkdir /tmp/devnet && pushd /tmp/devnet

./configure && make && make install

systemctl enable devnet
systemctl start devnet
```

Refer to the exhibit. A network engineer works on a Cisco DevNet application that is available for download on the Cisco DevNet website. The engineer prepares a bash script to automate operations that relate to the application. Which workflow is being automated in the bash script?

A compilation and configuration of the Cisco DevNet application so that it must be started manually

B compilation, installation, and configuration of the Cisco DevNet application so that it starts automatically ✓ CORRECT

C compilation and installation of the Cisco DevNet application for an autonomous deployment model

D installation and configuration of the Cisco DevNet application so that it starts manually

SUGGESTED ANSWER

The suggested answer is B.

The bash script provided performs several sequential steps to automate the workflow for the Cisco DevNet application. First, it downloads the application source code from a specified URL and saves it to a temporary directory. It then extracts the downloaded tar file and creates a new directory where the source code will reside. The script navigates to this newly created directory, then runs a series of commands to configure, compile, and install the application. Finally, the script enables the application to start automatically when the system boots and starts the application immediately. This comprehensive process includes compilation, installation, and configuration steps, and ensures that the application will start automatically, fulfilling all the necessary criteria listed in option B.

QUESTION 150

Issued to kamjunhao@gmail.com · October 17, 2026

A company wants to accelerate the adoption of technologies that will simplify its operations. Staff have different skill levels and are looking for a single site to find easy-to-consume beginner, intermediate, and advanced multidomain use cases. Which site should be used?

A DevNet Ecosystem Exchange

B DevNet Automation Exchange ✓ CORRECT

C DevNet Creations

D DevNet Support Forum

SUGGESTED ANSWER

The suggested answer is B.

The DevNet Automation Exchange is the site designed to provide a comprehensive library of use cases categorized by skill level, such as beginner, intermediate, and advanced. This site helps users quickly find practical code examples and solutions applicable to various domains, thereby facilitating the adoption of technology to simplify operations.

Issued to kamjunhao@gmail.com · October 17, 2026

Which Cisco product provides malware and content filtering through DNS?

A Cisco ASA Firepower module

B Cisco AMP

C Cisco ISE

D Cisco Umbrella ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is D.

Cisco Umbrella provides malware and content filtering through DNS. It processes billions of DNS requests per day, analyzes and learns about various activities, and blocks requests to malicious or unwanted destinations before a connection is established. Its services include blocking malware, ransomware, and phishing attempts from malicious or fraudulent sites.

Issued to kamjunhao@gmail.com · October 17, 2026

Which Cisco platform provides organization-wide automation, security, policy enforcement, and agility across wired and wireless networks?

A Cisco SD-WAN

B Cisco DNA Center ✓ CORRECT

C Cisco ACI

D Cisco Umbrella

SUGGESTED ANSWER

The suggested answer is B.

Cisco DNA Center is the platform that provides organization-wide automation, security, policy enforcement, and agility across wired and wireless networks. It serves as a centralized management dashboard for controller-based automation, network assurance, and security, making it the correct choice for comprehensive network management within an organization.

Issued to kamjunhao@gmail.com · October 17, 2026

What is a characteristic of the Cisco Finesse platform?

A Applications allow services to be invoked on a network triggered event. ✓ CORRECT

B The platform provides a ready-to-go platform for HD video and audio conferencing.

C Applications are added to the platform from the desktop remotely.

D The platform includes an enterprise-class IM engine.

SUGGESTED ANSWER

The suggested answer is A.

A characteristic of the Cisco Finesse platform is that applications allow services to be invoked based on network-triggered events. Cisco Finesse is a flexible, open development platform designed to provide custom applications and integrations for contact centers, allowing real-time reactions to specific events within the network, such as incoming calls or changes in customer account statuses.

Issued to kamjunhao@gmail.com · October 17, 2026

An engineer needs to retrieve a list of locally available Firepower Threat Defense upgrade packages by using the Firepower Management Center REST API. The engineer uses a GET request with the URL:

/api/fmc_platform/v1/updates/upgradepackages/f413afeb-e6f6-75f4-9169-6d9bd49s625eWhat does `f413afeb-e6f6-75f4-9169-6d9bd49s625e` represent?

A container UUID

B package UUID ✓ CORRECT

C domain UUID

D object UUID

SUGGESTED ANSWER

The suggested answer is B.

The UUID provided in the URL identifies a specific package. The URL pattern used for accessing details about specific upgrade packages in the Firepower Management Center REST API includes the package UUID. This identifier helps to fetch the details of a particular upgrade package from the list of locally available Firepower Threat Defense upgrade packages.

Refer to the exhibit.

```
docker run -p 3000:5000 f877cbf2ff33
```

A process running on the host wants to access the service running inside this Docker container. Which port or ports are used to make a connection?

A only outbound connections between 3000 and 5000 are possible

B port 3000 ✓ CORRECT

C any port between 3000 and 5000

D port 5000

SUGGESTED ANSWER

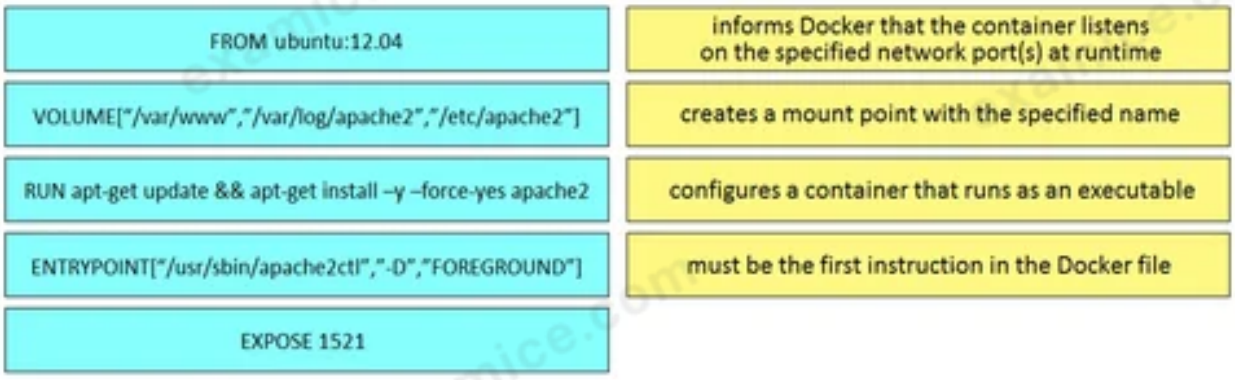
The suggested answer is B.

The correct port that a process running on the host uses to access the service running inside the Docker container is port 3000. The command 'docker run -p 3000:5000' maps port 3000 on the host to port 5000 in the container. This means that connections to port 3000 on the host are forwarded to port 5000 in the Docker container. Therefore, the process on the host will use port 3000 to make the connection, allowing it to access the service running inside the container on port 5000.

DRAG DROP -

Drag and drop the Dockerfile instructions from the left onto the descriptions on the right. Not all options are used.

Select and Place:



SUGGESTED ANSWER

Issued to kamjunhao@gmail.com · October 17, 2026

What are two security benefits of a Docker-based application? (Choose two.)

A natively secures access to secrets that are used by the running application

B guarantees container images are secured and free of vulnerabilities

C easier to patch because Docker containers include only dependencies that the application requires ✓ CORRECT

D prevents information leakage occurring when unhandled exceptions are returned in HTTP responses

E allows for separation of applications that traditionally run on the same host ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is C, E.

Two security benefits of a Docker-based application are that it is easier to patch because Docker containers include only dependencies that the application requires, and it allows for separation of applications that traditionally run on the same host. Containers streamline the updating process as dependencies are minimal and specific to the application, reducing the risk of vulnerabilities. Furthermore, containerization isolates applications from each other even if they share the same host, enhancing security by preventing potential interference and minimizing the attack surface.

Issued to kamjunhao@gmail.com · October 17, 2026

A local Docker alpine image has been created with an image ID of a391665405fe and tagged as `latest`. Which command creates a running container based on the tagged image, with the container port 8080 bound to port 80 on the host?

A docker build -p 8080:80 alpine:latest

B docker exec -p 8080:80 alpine:latest

C docker start -p 8080:80 alpine:latest

D docker run -p 8080:80 alpine:latest ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is D.

To create a running container based on the tagged image, with the container port 8080 bound to port 80 on the host, you should use the 'docker run' command. The correct syntax is 'docker run -p hostPort:containerPort imageTag'. Therefore, the command 'docker run -p 8080:80 alpine:latest' creates a new, running container from the 'alpine:latest' image, mapping port 8080 on the host to port 80 within the container.

Refer to the exhibit.

```
# base image
FROM alpine

# Install python and pip
RUN apk add --update py2-pip

# Install python modules needed by the python app
COPY requirements.txt /usr/src/app/
RUN pip install --no-cache-dir -r /usr/src/app/requirements.txt

#copy local files for the app to run
COPY app.py /usr/src/app/
COPY templates/index.html /usr/src/app/templates/

EXPOSE 8080

#run the application
CMD ["python", "/usr/src/app/app.py"]
```

Which function does the **EXPOSE** line perform when building an image from this Dockerfile?

A Local port 8080 of the container that launched with this docker image is accessible from co-hosted containers and external clients.

B Local port 8080 is not reachable at all because no IP address is specified along with the EXPOSE command of the Dockerfile.

C Local port 8080 is accessible from other containers running on the same docker host as the container that launched with this docker image. ✓ CORRECT

D Local port 8080 of the container that launched with this docker image is accessible to external clients.

SUGGESTED ANSWER

The suggested answer is C.

The EXPOSE instruction in a Dockerfile informs Docker that the container listens on the specified network port at runtime. However, by default, the EXPOSE instruction does not publish the port, meaning it does not make the port accessible from the host or external

clients. It only makes the stated ports available for inter-container communication on the same Docker host. To make the port accessible to external clients or co-hosted containers, additional steps such as using the -p or -P flags during the docker run command to publish and map the ports are required.

QUESTION 160

Which two elements are foundational principles of DevOps? (Choose two.)

A organizing cross-functional teams over organizational silos ✓ CORRECT

B designing applications as microservices

C encouraging containers for the deployment of applications

D automating over documenting ✓ CORRECT

E optimizing the cost of infrastructures

SUGGESTED ANSWER

The suggested answer is A, D.

Organizing cross-functional teams over organizational silos emphasizes collaboration and breaking down barriers between departments, which is a key principle of DevOps. Automating processes over extensive documentation is another fundamental principle of DevOps, as it focuses on enhancing efficiency and reducing manual interventions through automation. These elements promote smoother and quicker development and deployment cycles, which are core objectives of DevOps.

Issued to kamjunhao@gmail.com · October 17, 2026

Which platform is run directly on top of a hypervisor?

A bare metal systems

B virtual machines ✓ CORRECT

C containers

D applications

SUGGESTED ANSWER

The suggested answer is B.

A hypervisor is specialized software that creates and runs virtual machines. Virtual machines are hosted directly on top of a hypervisor, which manages the hardware resources and allocates them to each virtual machine. Therefore, the correct answer is virtual machines.

Issued to kamjunhao@gmail.com · October 17, 2026

What are two characteristics of Bare Metal environments that are related to application deployment? (Choose two.)

A specifically designed for container-based workloads

B suitable for legacy applications that do not support virtualization ✓ CORRECT

C provides workloads with access to hardware features ✓ CORRECT

D not compatible with other cloud services such as PaaS or SaaS offerings

E provides the hypervisor to host virtual servers

SUGGESTED ANSWER

The suggested answer is B, C.

Bare Metal environments are suitable for legacy applications that do not support virtualization because such setups allow the software to run directly on the hardware without the need for a hypervisor layer. Additionally, they provide workloads with access to hardware features, ensuring applications can utilize the full capabilities of the hardware, which is often crucial for performance-intensive tasks.

Issued to kamjunhao@gmail.com · October 17, 2026

A new application is being developed that must be hardware independent. The application includes an administrative component which is accessed using a Windows desktop GUI. Installation and management of the environment must be fully automated. Which application deployment type meets the requirements?

A virtual machine ✓ CORRECT

B virtual Python environment

C container

D bare metal

SUGGESTED ANSWER

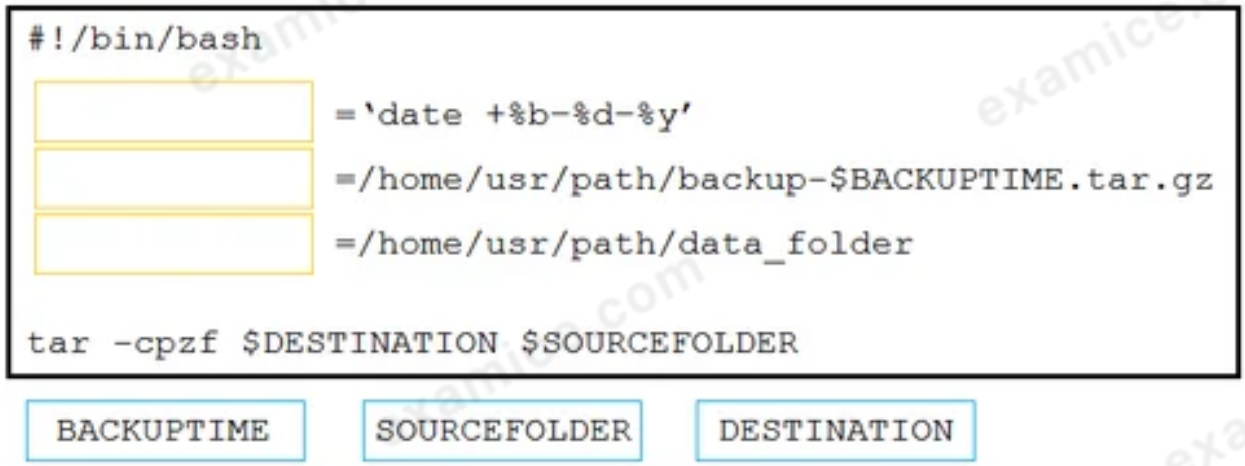
The suggested answer is A.

The application is required to be hardware independent and includes a Windows desktop GUI, which is accessed administratively. Given these requirements, a virtual machine is the most appropriate deployment type. Virtual machines are hardware-independent and can support a full Windows desktop GUI environment, allowing for the necessary installation and management automation. Containers generally do not support full graphic user interfaces effectively, especially for Windows desktop applications, limiting their suitability for this scenario.

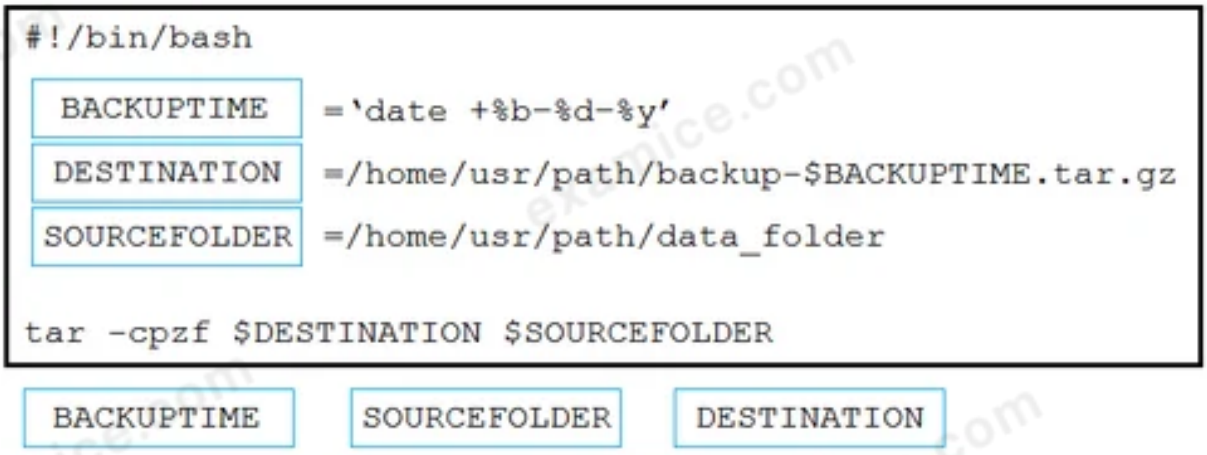
DRAG DROP -

Drag and drop the code from the bottom onto the box where the code is missing in the Bash script to complete the missing assignments.

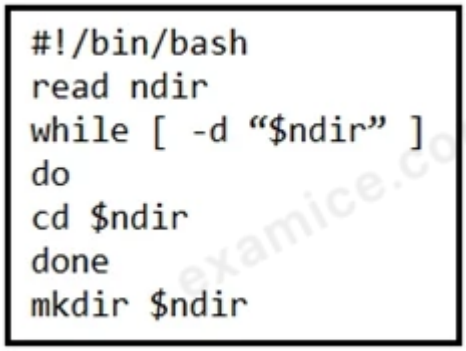
Select and Place:



SUGGESTED ANSWER



Refer to the exhibit.



```
#!/bin/bash
read ndir
while [ -d "$ndir" ]
do
cd $ndir
done
mkdir $ndir
```

What is the action of the Bash script that is shown?

A For all directories in the current folder, the script goes into the directory and makes a new directory.

B The script waits until input is entered. If the directory exists, the script goes into it until there is no directory with the same name, then it creates a new directory. ✓ CORRECT

C The script waits until input is entered, then it goes into the directory entered and creates a new directory with the same name.

D The script goes into the directory called ɹ€$ndirɹ€ and makes a new directory called ɹ€$ndirɹ€.

SUGGESTED ANSWER

The suggested answer is B.

The script waits for the user to input a directory name and assigns it to the variable 'ndir'. It then checks if a directory with that name exists using a while loop. If the directory exists, the script changes to that directory and checks again. This loop continues until a directory with that name does not exist. Finally, it creates a new directory with the name given as input.

Issued to kamjunhao@gmail.com · October 17, 2026

When using the Bash shell, how is the output of the devnet command saved to a file named `output.txt`?

A devnet > output.txt ✓ CORRECT

B devnet | output.txt

C devnet < output.txt

D devnet & output.txt

SUGGESTED ANSWER

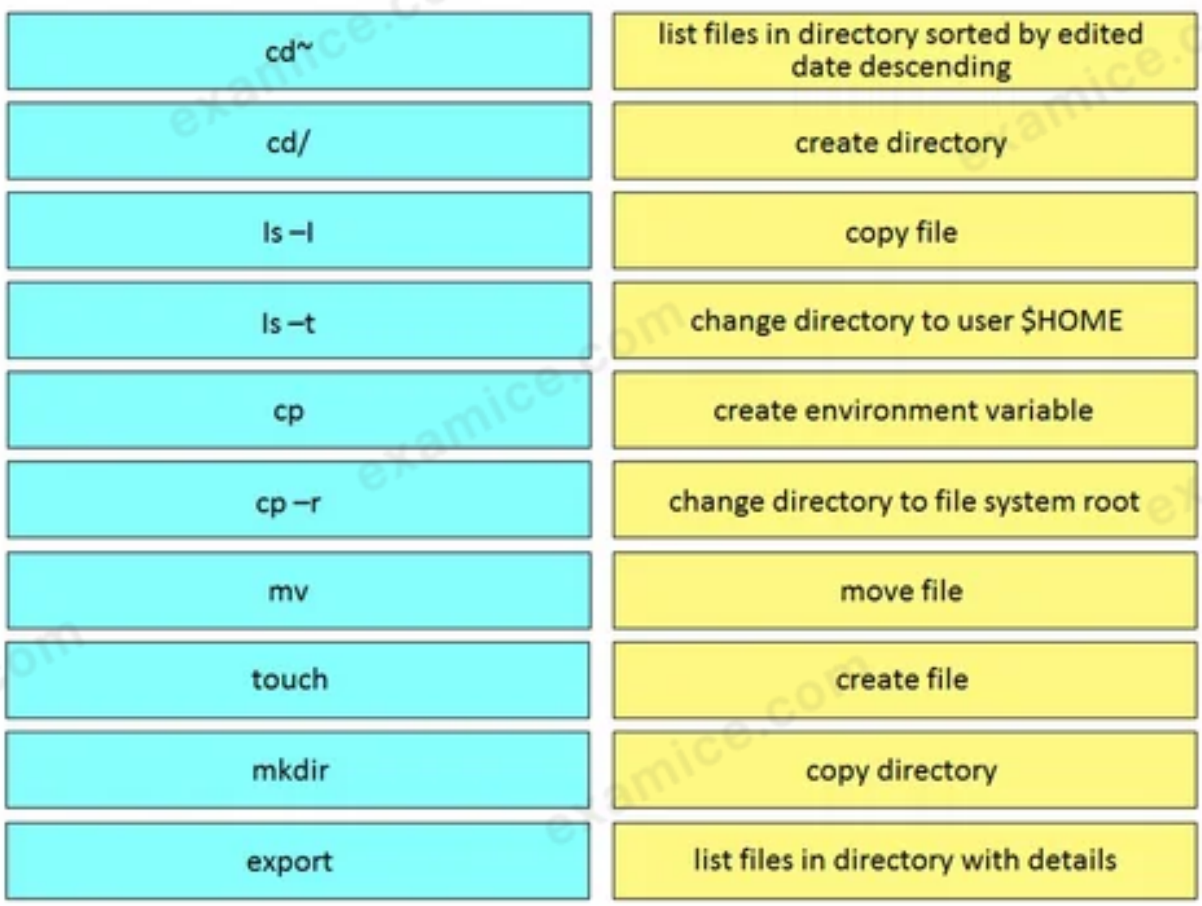
The suggested answer is A.

To save the output of the 'devnet' command to a file named 'output.txt' in Bash, the correct syntax is 'devnet > output.txt'. The '>' operator redirects the standard output of the command to the specified file, creating or overwriting it if it already exists.

DRAG DROP -

Drag and drop the Bash commands from the left onto the related actions on the right.

Select and Place:



SUGGESTED ANSWER

Personal copy · kamjunhao@gmail.com

FILL BLANK -Fill in the blanks to complete the Bash script in which each file in a directory is renamed to its SHA256 hash?

TARGET_DIR=/usr/local/certs -for f in '_____ _____'; do_____ $f' openssl sha -sha256 -r < $f | cut -f1 -cl' ''

SUGGESTED ANSWER

To complete the Bash script that renames each file in a directory to its SHA256 hash, you can use the following approach: First, the `TARGET_DIR` variable is set to the desired directory, and then a for loop iterates over each file in that directory. Within the loop, the file is hashed using `openssl` and renamed accordingly. Here's the complete script: TARGET_DIR=/usr/local/certs for f in $(ls $TARGET_DIR); do mv $TARGET_DIR/$f $TARGET_DIR/$(openssl sha256 -r < $TARGET_DIR/$f | cut -d ' ' -f 1) done Explanation: 1. `TARGET_DIR=/usr/local/certs`: Sets the target directory. 2. `for f in $(ls $TARGET_DIR); do`: Loops through each file in the target directory. 3. `mv $TARGET_DIR/$f $TARGET_DIR/$(openssl sha256 -r < $TARGET_DIR/$f | cut -d ' ' -f 1)`: Moves (renames) the file to its SHA256 hash. The script lists the files in the directory and renames each one to its hashed value.

Issued to kamjunhao@gmail.com · October 17, 2026

After a project is migrated to a new codebase, the `old_project` directory must be deleted. This directory has multiple read-only flies, and it must be deleted recursively without prompting for confirmation. Which Bash command must be used?

A rmdir -p old_project

B rm -rf old_project ✓ CORRECT

C rm -r old_project

D rmdir old_project

SUGGESTED ANSWER

The suggested answer is B.

To delete a directory and its contents recursively without prompting for confirmation, especially when dealing with read-only files, the `rm` command with both the `-r` (recursive) and the `-f` (force) options should be used. Using `rm -rf old_project` will ensure that the entire directory, `old_project`, along with all its files and subdirectories, is deleted without any prompts or permissions issues.

What is a benefit of using edge computing in an IoT implementation?

A high speed in data processing ✓ CORRECT

B low network design complexity

C low cost in network design

D high availability for network components

SUGGESTED ANSWER

The suggested answer is A.

A benefit of using edge computing in an IoT implementation is high speed in data processing. Edge computing brings computation and data storage closer to the location where it is needed, which reduces latency and allows for quicker data processing. This is especially important for IoT devices that need to respond to data in real-time or near real-time, significantly improving the efficiency and response times of the system.

examice.com

What is the benefit of edge computing?

A It reduces network latency by moving processing closer to the data source. ✓ CORRECT

B It reduces data velocity from devices or other data sources.

C It simplifies security as devices and processing are brought closer together.

D It removes the need for centralized data processing.

SUGGESTED ANSWER

The suggested answer is A.

Edge computing reduces network latency by moving processing closer to the data source. By processing data near where it is generated, edge computing minimizes the time and distance required for data to travel to a centralized data center or cloud, thereby reducing latency. This results in quicker response times and more efficient data handling, which is the primary benefit of edge computing.

examice.com

DRAG DROP -

Refer to the exhibit.

```
#!/bin/bash
<item 1>='date +%b-%d-%y'

<item 2>=/home/usr/path/backup-$BACKUPTIME.tar.gz

<item 3>=/home/usr/path/data_folder

tar -cpzf $DESTINATION $SOURCEFOLDER
```

Drag and drop the variables from the left onto the item numbers on the right that match the missing assignments in the exhibit.

Select and Place:

BACKUPTIME	<item 1>
SOURCEFOLDER	<item 2>
DESTINATION	<item 3>

SUGGESTED ANSWER

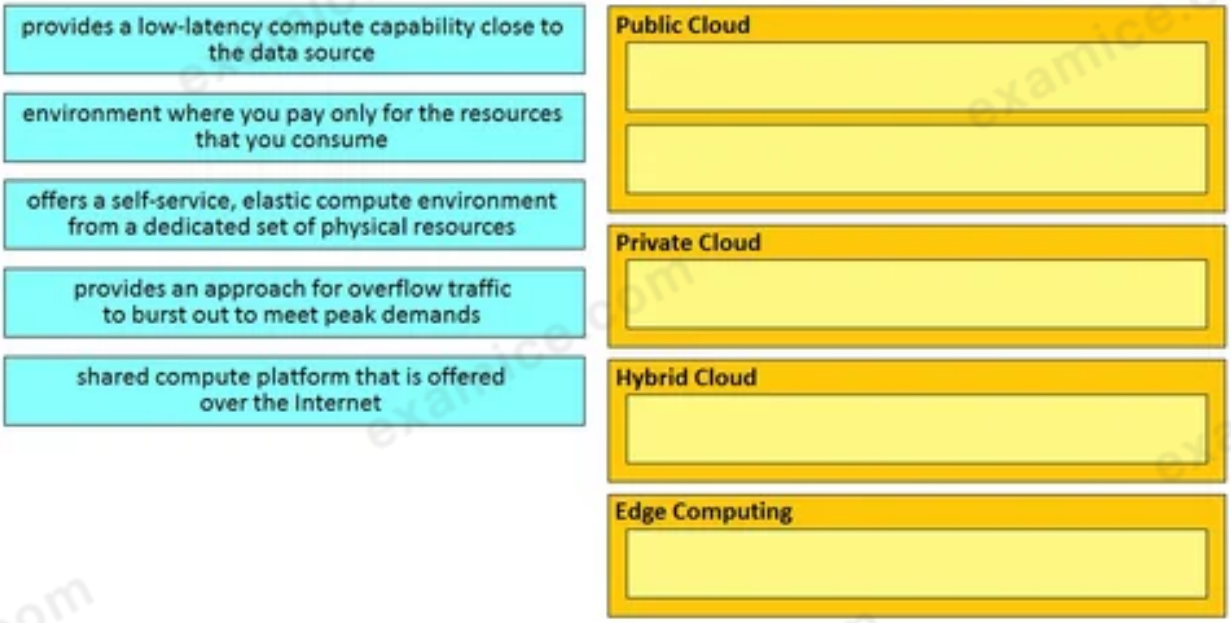
Reference:
https://medium.com/@fotios.floros/linux-backup-script-1722cc9c2bf6
(use tar in order to create your backup file)

BACKUPTIME	BACKUPTIME
SOURCEFOLDER	DESTINATION
DESTINATION	SOURCEFOLDER

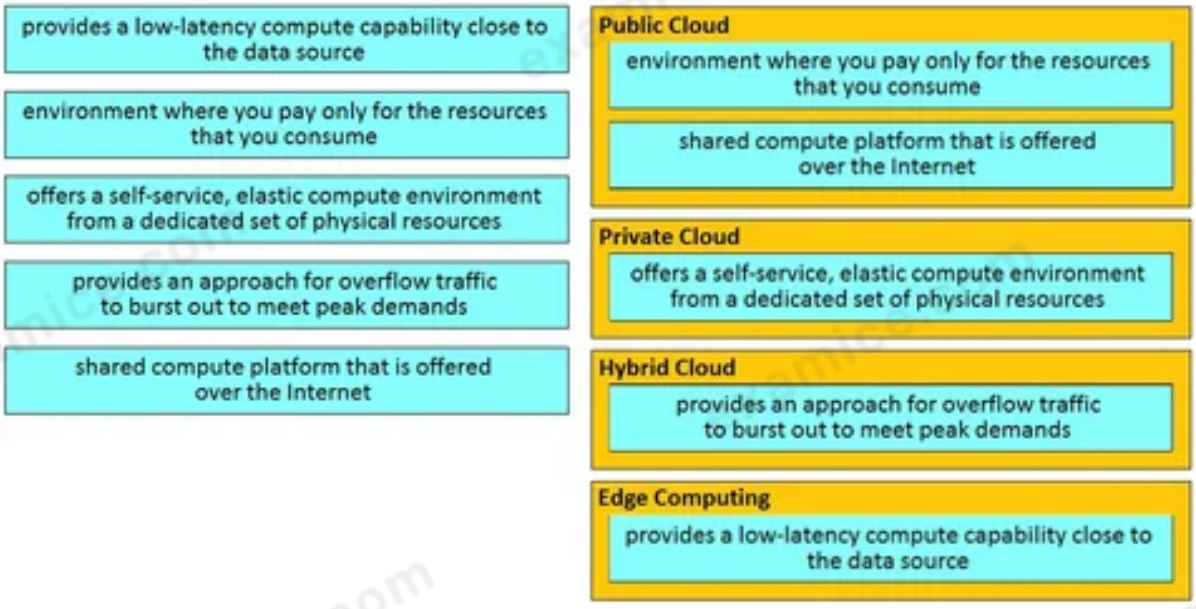
DRAG DROP -

Drag and drop the descriptions from the left onto the correct application deployment models on the right.

Select and Place:



SUGGESTED ANSWER



Issued to kamjunhao@gmail.com · October 17, 2026

Which type of threat occurs when an attacker can send hostile data to an interpreter within an application?

A sensitive data exposure

B broken authentication

C cross-site scripting

D injection ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is D.

An injection threat occurs when an attacker can send hostile data to an interpreter within an application. This can include SQL, NoSQL, OS, and LDAP injections. Injection flaws are among the most common and dangerous vulnerabilities, as they allow attackers to execute arbitrary commands or access data without proper authorization.

Issued to kamjunhao@gmail.com · October 17, 2026

A company is looking for a cloud deployment which will only use the on-premise infrastructure, is user self-service, and is **easy** to scale. Which cloud solution should be selected for these requirements?

A multi

B private ✓ CORRECT

C hybrid

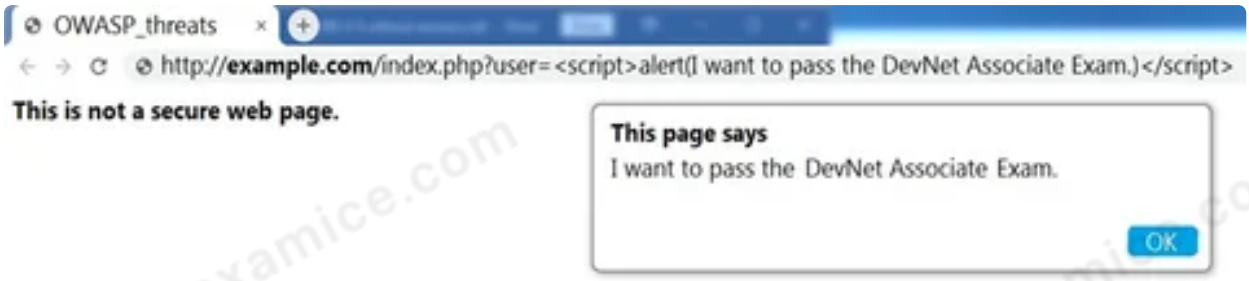
D public

SUGGESTED ANSWER

The suggested answer is B.

The best cloud solution for a company that wants to use only on-premise infrastructure while providing user self-service and easy scalability is a private cloud. A private cloud is hosted on-premises and offers the control and security of on-premise traditional infrastructure. It allows for user self-service within the infrastructure owned by the company, which aligns with the requirement of using only on-premise resources.

Refer to the exhibit.



Which OWASP threat does this example demonstrate?

A broken access control

B cross-site scripting ✓ CORRECT

C SQL injection

D man-in-the-middle

SUGGESTED ANSWER

The suggested answer is B.

The exhibit shows a URL where a script is injected directly into the web application's URL, leading to an alert message being displayed on the page. This is a classic example of cross-site scripting (XSS). In XSS attacks, malicious scripts are injected into otherwise benign and trusted websites. The primary goal is to execute scripts in the context of the victim's browser, which can lead to various malicious outcomes such as stealing cookies, session tokens, or other sensitive data. The other options, such as broken access control, SQL injection, and man-in-the-middle, do not match the scenario depicted in the exhibit.

Issued to kamjunhao@gmail.com · October 17, 2026

What should a CI/CD pipeline aim to achieve?

A to require minimal manual interaction ✓ CORRECT

B to allow manual testing before deployment

C to support new deployments on a fixed monthly schedule

D to provide a documented process for feedback

SUGGESTED ANSWER

The suggested answer is A.

A CI/CD pipeline should aim to require minimal manual interaction. The primary goal of CI/CD (Continuous Integration/Continuous Deployment) is to automate the software development process, from integration and testing to deployment, to ensure rapid and reliable delivery of software updates. Manual intervention is intended to be minimized to reduce errors and improve efficiency.

A function my_func() returns True when it executes normally. Which Python snippet tests my_func()?

A.

```
def test_func(self):
    self.assertTrue(my_func())
```

B.

```
def test_func(self):
    self.assertFalse(my_func())
```

C.

```
def test_func(self):
    self.assertEqual(my_func(),'{true}')
```

D.

```
def test_func(self):
    self.assertRaises(my_func())
```

SUGGESTED ANSWER

Reference:
https://github.com/django/django/blob/master/tests/auth_tests/test_mixins.py

Issued to kamjunhao@gmail.com · October 17, 2026

What are two roles of an artifact repository in a CI/CD pipeline? (Choose two.)

A stores files needed and generated during the build process ✓ CORRECT

B provides traceable, searchable, and manageable binary files ✓ CORRECT

C is required for CI/CD pipelines executed on a public cloud infrastructure

D allows for comparing and merging changes in the source code

E is required for managing open source software

SUGGESTED ANSWER

The suggested answer is A, B.

An artifact repository in a CI/CD pipeline stores files needed and generated during the build process, making them easily accessible for future stages. It also provides traceable, searchable, and manageable binary files, ensuring that artifacts can be reliably used, versioned, and reproduced when necessary. These roles are crucial for maintaining a streamlined and efficient CI/CD pipeline.

Issued to kamjunhao@gmail.com · October 17, 2026

A company is adopting DevOps as part of an internal transformation, and is reviewing the success of the first deployments. Developers and engineers are working together to resolve any resulting issues. However, this new way of working has increased overhead, and the team is finding it difficult to complete releases in time.

Which area of the CALMS framework must the company target for improvement?

A Collaboration

B Lean ✓ CORRECT

C Sharing

D Measurement

SUGGESTED ANSWER

The suggested answer is B.

Lean principles are focused on maximizing efficiency and minimizing waste, which is crucial when the team is struggling to complete releases on time due to increased overhead. By targeting Lean practices, the company can streamline processes, reduce unnecessary work, and improve the speed and quality of software development and delivery.

```
FROM nginx

ARG DELAY="30"
ENV INITIAL_DELAY=${DELAY}

ENTRYPOINT ["/bin/sh"]

CMD ["-c", "/bin/sleep ${INITIAL_DELAY} && nginx -g
'daemon off;'"]
```

Refer to the exhibit. Which command, including arguments, is executed when the container starts?

A /bin/sh -c "/bin/sleep 30 && nginx -g 'daemon off;'" ✓ CORRECT

B /bin/sh -c "/bin/sleep 10 && nginx -g 'daemon off;'"

C /bin/bash -c "/bin/sleep 30 && nginx -g 'daemon off;'"

D /bin/sh -c "/bin/sleep 30" && nginx -g 'daemon off;'

SUGGESTED ANSWER

The suggested answer is A.

The Dockerfile specifies an ARG named DELAY with a default value of 30. This value is assigned to the environment variable INITIAL_DELAY. The ENTRYPOINT is set to /bin/sh, and the CMD specifies the arguments to be passed to /bin/sh when the container starts. The CMD uses /bin/sh -c to run the command /bin/sleep 30 && nginx -g 'daemon off;'. This matches the option that correctly reflects this command execution path.

Issued to kamjunhao@gmail.com · October 17, 2026

A developer created a new secure API and is testing the functionality locally. The API was added to a container for further testing, and other team members and applications need to connect to the API. Which command exposes the API in the container as port 8443 and allows connections external to the host?

A docker run -p 127.0.0.1:80:8443/tcp myapiservice

B docker run -p 0.0.0.0:443:8443/tcp myapiservice ✓ CORRECT

C docker run -p 0.0.0.0:8443:443/tcp myapiservice

D docker run -p 127.0.0.1:443:8443/tcp myapiservice

SUGGESTED ANSWER

The suggested answer is B.

To expose the API in the container as port 8443 and allow connections external to the host, the correct syntax is to use the -p flag with the host port and container port. The host port should be 8443, and the container port should be 8443. Therefore, the correct command is: docker run -p 0.0.0.0:8443:8443/tcp myapiservice. This ensures that the API inside the container, running on port 8443, is accessible externally on port 8443.

A developer has created a new image to use in a Docker build and has added a tag for the image by using the command:

```
$ docker tag 84fe411926287 local/app:0.4
```

Which command must be executed next to build the Docker image using the tag?

A $ docker build -p local/app:0.4

B $ docker run -t local/app:0.4

C $ docker run -p local/app:0.4

D $ docker build -t local/app:0.4 ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is D.

To build a Docker image and assign a tag to it, you use the 'docker build' command with the '-t' option followed by the tag name. The '-t' signifies 'tag' in the Docker build procedure. The proper command syntax to build the Docker image using the tag 'local/app:0.4' is 'docker build -t local/app:0.4'. This command will create a new image and tag it with the specified tag.

Issued to kamjunhao@gmail.com · October 17, 2026

Access to the management interface of devices must be restricted by using SSH and HTTPS. Which two ports must be included in the ACLs for the protocols to work? (Choose two.)

A 22 ✓ CORRECT

B 23

C 80

D 443 ✓ CORRECT

E 880

SUGGESTED ANSWER

The suggested answer is A, D.

SSH (Secure Shell) operates on port 22, providing a secure channel over an unsecured network, while HTTPS (HyperText Transfer Protocol Secure) operates on port 443, ensuring secure communication over the internet. To restrict access to the management interface of devices using these protocols, the ACLs must include ports 22 and 443.

Issued to kamjunhao@gmail.com · October 17, 2026

What is a benefit of using a code review process in application development?

A accelerates the deployment of new features in an existing application

B provides version control during code development

C enables the quick deployment of new code

D eliminates common mistakes during development ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is D.

Using a code review process during application development helps eliminate common mistakes. This process involves multiple developers examining code to identify and correct errors, improve code quality, and ensure adherence to best practices and coding standards.

Issued to kamjunhao@gmail.com · October 17, 2026

Which OWASP threat takes advantage of a web application that the user has already authenticated to execute an attack?

A phishing

B DoS

C brute force attack

D CSRF ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is D.

Cross-Site Request Forgery (CSRF) is an attack that tricks a user into executing unwanted actions on a web application where they are currently authenticated. CSRF exploits the trust that a web application has in the user's browser, and it requires the user to be authenticated so that the attack can force the user to perform actions without their consent.

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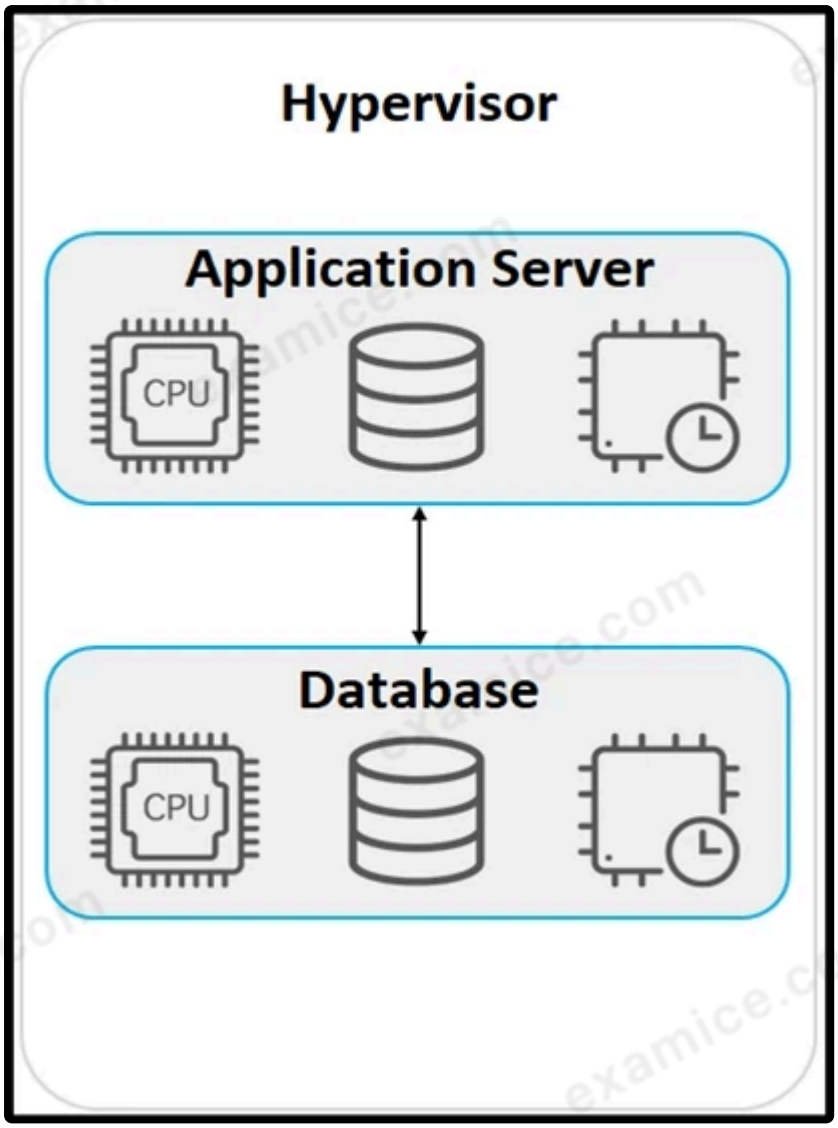

Refer to the exhibit. An application must be deployed on a server that has other applications installed. The server resources are to be allocated based on the deployment requirements. The OS for each application must be independent. Which type of deployment is used?

A hybrid

B virtual machines ✓ CORRECT

C containers

D bare metal

SUGGESTED ANSWER

The suggested answer is B.

The described scenario requires that the server resources be allocated based on deployment requirements and that the operating system for each application must be independent. This requirement precisely matches the characteristics of virtual machines. Virtual machines provide isolated environments by running separate instances of operating systems on the same physical server, managed by a hypervisor. This isolation ensures that each application operates independently with its own OS, fulfilling the deployment requirements stated in the question.

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What is a benefit of a distributed version control system?

A encourages users to commit small places of work to the system more frequently

B ensures that all code meets minimum standards before being committed to the system

C allows users to work on the codebase even when not connected to the Internet ✓ CORRECT

D ensures that all code is tested before being committed to the system

SUGGESTED ANSWER

The suggested answer is C.

A key benefit of a distributed version control system is that it allows users to work on the codebase even when not connected to the Internet. This is possible because each developer has a local copy of the entire repository on their machine. With this setup, users can commit, branch, and merge changes locally without needing to be connected to a central server. This contrasts with centralized version control systems, which require a connection to the main server to perform such actions.

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A company requires a new platform to store large volumes of log data that is generated in their assembly plant. The platform must be scalable, so the underlying technology must provide a path for fast deployment of new nodes for data and indexing? The data contains sensitive information, so specialist encryption tools must be used to secure the data. Which application deployment model meets the requirements?

A edge

B private cloud

C public cloud

D hybrid cloud ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is D.

A hybrid cloud deployment model meets the requirements by combining the security of a private cloud with the scalability of a public cloud. The private cloud ensures that sensitive data is encrypted and secure, while the public cloud allows for quick deployment of new nodes to handle large volumes of log data. This model leverages the strengths of both environments, making it a suitable choice for the given scenario.

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What is a procedural component of CI/CD pipelines?

A Bundled artifacts must be generated.

B Every test must pass for the pipeline to continue its execution. ✓ CORRECT

C Some tests fall without stopping the pipeline execution.

D The release label that identifies the build is optional.

SUGGESTED ANSWER

The suggested answer is B.

A key procedural component of CI/CD pipelines is ensuring that every test must pass for the pipeline to continue its execution. This step is critical as it helps maintain the integrity and quality of the code by preventing faulty code from proceeding further in the deployment process. If a test fails, the pipeline stops, and the issue must be resolved before continuing, which makes this a procedural requirement in CI/CD pipelines.

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Which OWASP threat occurs when untrusted data is supplied as input to an interpreter as part of a command or query?

A broken authentication

B data exposure

C injection ✓ CORRECT

D cross-site scripting

SUGGESTED ANSWER

The suggested answer is C.

The OWASP threat that occurs when untrusted data is supplied as input to an interpreter as part of a command or query is known as an injection. Injection flaws, such as SQL, NoSQL, OS, and LDAP injection, occur when untrusted data is sent to an interpreter. These flaws are very prevalent, notably in legacy applications, and can result in unauthorized access or data leaks.

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Which product provides network controller-level management features?

A Cisco ISE

B Cisco DNA Center ✓ CORRECT

C Cisco UCS Manager

D Cisco NX-OS

SUGGESTED ANSWER

The suggested answer is B.

Cisco DNA Center provides network controller-level management features. It offers a centralized interface to integrate management and automation of network infrastructure, making it the suitable choice for network controller management. Cisco ISE focuses on network access control and security, Cisco UCS Manager manages server hardware, and Cisco NX-OS is an operating system for network switches.

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What is the purpose of the Cisco VIRL software tool?

A to verify configurations against compliance standards

B to automate API workflows

C to simulate and model networks ✓ CORRECT

D to test performance of an application

SUGGESTED ANSWER

The suggested answer is C.

Cisco VIRL (Virtual Internet Routing Lab) software tool is specifically designed to simulate and model networks. It enables users to create and test virtual network environments, providing a valuable platform for learning and experimentation without the need for physical hardware.

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What are two advantages of the Model-View-Controller software design pattern? (Choose two.)

A allows for multiple views of the same model ✓ CORRECT

B separates responsibilities of the code, which makes future modifications easier ✓ CORRECT

C simplifies network automation

D makes code easier to deploy using CI/CD pipelines

E reduces the need for error handling

SUGGESTED ANSWER

The suggested answer is A, B.

The Model-View-Controller (MVC) software design pattern offers distinct advantages that make it an effective choice in software architecture. Firstly, it allows for multiple views of the same model, meaning the data and business logic (the model) can be represented in various ways without altering the underlying structure. This flexibility is critical in modern applications where the same data must be displayed differently across devices and interfaces. Secondly, MVC separates the responsibilities of the code into three interconnected components: the model, view, and controller. This separation of concerns enhances the maintainability of the application because changes in the user interface (view) or the business logic (model) can be made independently without affecting other components. This modularity also simplifies debugging and extends the lifetime of the software by making it easier to incorporate future modifications.

```
module ietf-ip {
 namespace "urn:ietf:params:xml:ns:yang:ietf-ip";
 prefix ip;
 import ietf-interfaces {
   prefix if;
 }
 augment "/if:interfaces/if:interface" {
   container ipv4 {
     leaf enabled {
       type boolean;
       default true;
     }
     list address {
       key "ip";
       leaf ip {
         type inet:ipv4-address-no-zone;
       }
       choice subnet {
         mandatory true;
         leaf prefix-length {
           type uint8 {
             range "0..32";
           }
         }
         leaf netmask {
           type yang:dotted-quad;
         }
       }
     }
   }
 }
}
```

```
module ietf-interfaces {
  namespace "urn:ietf:params:xml:ns:yang:ietf-
  interfaces";
  prefix if;
  typedef interface-ref {
    type leafref {
      path "/if:interfaces/if:interface/if:name";
    }
  }
  container interfaces {
    list interface {
      key "name";
     leaf name {
        type string;
      }
      leaf description {
        type string;
      }
      leaf type {
        type identityref {
          base interface-type;
        }
        mandatory true;
      }
      leaf enabled {
        type boolean;
        default "true";
      }
    }
  }
}
```

Refer to the exhibit. Which JSON snippet configures a new interface according to YANG model?

- A.

```
"ietf-interfaces": {
  interface": {
    "name": "Loopback100",
    "enabled": true,
    "ietf-ip": {
      ipv4": {
       "address": [
       {
         "ip": "10.255.254.1",
         "netmask": "255.255.255.0"
       } ] } }
}
```

- B.

```
" interface"  {
  "name": "Loopback100",
  "enabled": true,
  "ipv4": {
    "address": [
    {
      "ip": "10.255.254.1",
      "netmask": "255.255.255.0"
     } ] }
}
```

- C.

```
"ietf-interfaces:interface": {
  "name": "Loopback100",
  "enabled": true,
  "ietf-ip:ipv4:address": [
    {
      "ip": "10.255.254.1",
      "netmask": "255.255.255.0"
     } ]
}
```

- D.

```
"ietf-interfaces:interface": {
  "name": "Loopback100",
  "enabled": true,
  "ietf-ip:ipv4": {
    "address": [
    {
      "ip": "10.255.254.1",
      "netmask": "255.255.255.0"
     } ] }
}
```

SUGGESTED ANSWER

The correct JSON snippet that configures a new interface according to the YANG model is D. The provided YANG model specifies that the `interface` list is within the `ietf-interfaces:interfaces` container and the `ipv4` container is added under each interface using the `ietf-ip` namespace. JSON format D correctly represents these structures including namespace prefixes and hierarchical organization. The representation aligns with how containers, lists, and leaves are structured and referenced within the YANG modules. Therefore, the correct answer is D.

Refer to the exhibit.

```
module ex-ethernet {
     namespace "http://example.com/ethernet";
     prefix "eth";
     import ietf-interfaces {
       prefix if;
     }
     augment "/if:interfaces/if:interface" {
       when "if:type = 'ethernetCsmacd'";
       container ethernet {
         must "../if:location" {
           description
             "An ethernet interface must specify the physical location of the ethernet hardware.";
         }
         choice transmission-params {
           case auto {
             leaf auto-negotiate {
               type empty;
             }
           }
           case manual {
             leaf duplex {
               type enumeration {
                 enum "half";
                 enum "full";
               }
             }
             leaf speed {
               type enumeration {
                 enum "10Mb";
                 enum "100Mb";
                 enum "1Gb";
                 enum "10Gb";
               }
             }
           }
         } // other ethernet specific params...
       }
     }
```

What is represented in this YANG module?

A interface management ✓ CORRECT

B BGP

C OpenFlow

D topology

SUGGESTED ANSWER

The suggested answer is A.

The YANG module shown is designed for the management of Ethernet interfaces. This is evident from the key terms and structure seen in the module, such as 'augment', 'container ethernet', 'choice transmission-params', and various parameters specific to Ethernet interfaces like 'auto-negotiate', 'duplex', and 'speed'. These elements indicate that the module pertains to the configuration and management of network interfaces, specifically Ethernet interfaces.

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QUESTION 197

Issued to kamjunhao@gmail.com · October 17, 2026

Which CI/CD tool is an automation tool used to build, test, and deploy software?

A Git

B Gradle

C Nagios

D Jenkins ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is D.

Jenkins is an open-source automation server which is widely used for continuous integration and continuous delivery (CI/CD) purposes. It enables developers to automate the building, testing, and deploying of software, making it an essential tool in the CI/CD pipeline. Git is a version control system, Gradle is a build automation tool, and Nagios is used for network monitoring, none of which are primarily focused on the CI/CD pipeline automation.

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Issued to kamjunhao@gmail.com · October 17, 2026

An automation script fails to connect to an internal server exactly 1 out of 2 times it is executed. This behavior is seen from different clients. Which networking device must be at fault?

A load balancer ✓ CORRECT

B laptop on which the script is running

C switch

D router

SUGGESTED ANSWER

The suggested answer is A.

An intermittent connectivity issue that occurs exactly 1 out of 2 times suggests the problem may be related to a device responsible for balancing or distributing connections. A load balancer distributes network or application traffic across multiple servers to ensure no single server becomes overwhelmed. If one server in the pool is failing or misconfigured, it can result in a failure precisely half the time as attempts alternate between the working and non-working server. Therefore, the load balancer is the most logical culprit for this scenario.

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What are two advantages of YANG-based approaches for infrastructure automation? (Choose two.)

A multi-platform vendor abstraction ✓ CORRECT

B compiles to executables that run on network devices

C designed to reflect networking concepts ✓ CORRECT

D directly maps to JavaScript

E command line driven interface

SUGGESTED ANSWER

The suggested answer is A, C.

YANG-based approaches for infrastructure automation have several advantages. One advantage is multi-platform vendor abstraction, which allows for a standardized way of representing data across different vendors' equipment, making automation easier and more consistent. Another advantage is that YANG is designed to reflect networking concepts, meaning that it is inherently suited for describing network configurations and operations, making it a powerful tool for network automation.

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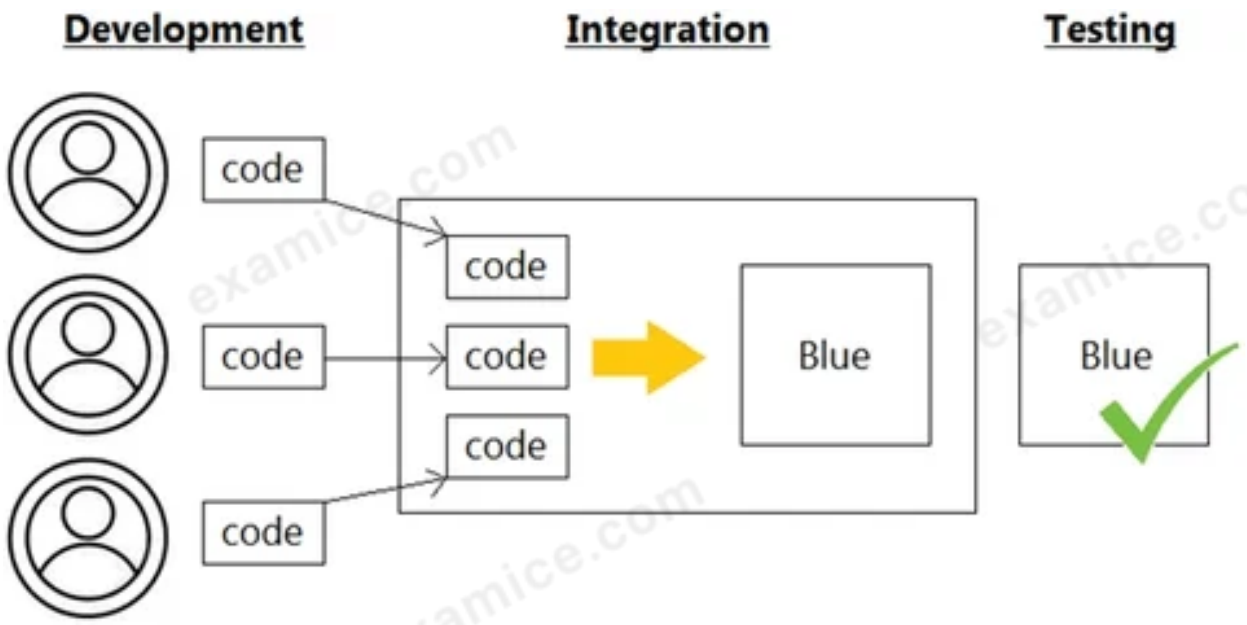
Refer to the exhibit.



Which infrastructure automation method is shown?

A CI/CD pipeline ✓ CORRECT

B Lean

C Agile

D Waterfall

SUGGESTED ANSWER

The suggested answer is A.

The exhibit shows a process where code from multiple developers is integrated into a single shared repository and then tested as a cohesive unit, which aligns with the principles of a CI/CD pipeline. Continuous Integration (CI) involves developers frequently merging their code changes into a central repository, followed by automated builds and tests. Continuous Delivery (CD) extends CI by automating the release process, although this step is not explicitly shown in the exhibit. The process is focused on automating integration and testing, which is indicative of a CI/CD pipeline and not Lean, Agile, or Waterfall methodologies.

QUESTION 201

Issued to kamjunhao@gmail.com · October 17, 2026

```
#!/bin/bash
apt install nginx
export AVAILABLE_SITES_DIR = '/etc/nginx/sites-available'
export ENABLED_SITES_DIR = '/etc/nginx/sites_enabled'
cd /opt/nginx
cat sites.txt | while read site;
    do
        cp template.conf "$AVAILABLE_SITES_DIR/$site";
        ln -s "$AVAILABLE_SITES_DIR/$site"" $ENABLED_SITES_DIR';
        chown www-data:www-data "$AVAILABLE_SITES_DIR/$site";
    done
```

Refer to the exhibit. An engineer must configure a load balancer server. The engineer prepares a script to automate workflow by using Bash. The script installs the nginx package, moves to the /opt/nginx directory, and reads the sites.txt file for further processing. Based on the script workflow, which process is being automated within the loop by using the information in sites.txt?

A creating a file per each line in sites.txt with the information in template.conf, creating a link for the previously created file, and then changing the ownership of the created files ✓ CORRECT

B using the content of the file to create the template.conf file, creating a link from the created file to the /etc/nginx/sites_enabled, and then changing the file execution permissions

C creating a new file based on template.conf in the /etc/nginx/sites_enabled directory for each line in the sites.txt file, and then changing the file execution permissions

D using the information in the file to create a set of empty files in the /etc/nginx/sites_enabled directory, and then assigning the owner of the file

SUGGESTED ANSWER

The suggested answer is A.

The script accomplishes the following actions within the loop: it creates a file in the /etc/nginx/sites-available directory using the template.conf file for each line read from sites.txt. Next, it creates symbolic links for these newly created files in the /etc/nginx/sites-enabled directory. Finally, it changes the ownership of these files to the www-data user and group. This matches exactly with option A, which states creating a file per each line in